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Large-scale biodiversity intactness estimation using Bayesian hierarchical models

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Abstract

Biodiversity is declining globally, with abundance of native species reduced by at least 20% since the early 1900s, driven by land-use change, pollution and climate change. In response, several global-scale biodiversity models have been developed to evaluate the state of ecosystems. One example is the Biodiversity Intactness Index (BII), quantified using the PREDICTS biodiversity database. The BII is used as a core indicator for global biodiversity targets, but despite this has not been systematically evaluated in terms of predictive performance. In this study, I implement and test the BII species abundance model on out-of-sample data. This linear mixed model has a random effects structure reflecting the structure of the PREDICTS data used for training. This is benchmarked against a Bayesian hierarchical model with a structure based on fundamental biogeographic factors, and with additional bioclimatic and topographic covariates. The Bayesian hierarchical model achieves a 33% lower mean absolute error and 52% lower median absolute error compared to the original BII model. Calibration of this model, using a simple polynomial regression model, leads to improvements of 58% and 68%, respectively. The BII model exhibits a very limited linear relationship between predicted and observed values, mainly driven by the random effects structure that does not generalize well to new data.

Due to the BII definition of relative biodiversity intactness, these improvements do not translate one-to-one into improvements in the final index outputs. Those outputs are obtained by relating each prediction to an undisturbed reference site in a post-processing step. On average, the predicted abundance values of the hierarchical base model (before calibration) are 26% lower, and the final index outputs are 21% lower, than the BII baseline model. I also find that this scaling method leads to a problematic distribution of output values, with a large number of outputs outside the 0-1 index range, that need to be capped. In summary, the BII predictive performance and output scaling suggest caution in its use for assessment, policy and decision-making. This study demonstrates the effectiveness of Bayesian hierarchical models, but also that there is significant improvement potential in global biodiversity modeling and systematic testing of such models.

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1 Introduction

The motivation and background for this study is presented in Section 1.1, followed by a description of research aims, questions and delimitations in Section 1.2.

1.1 Setting and motivation

Biodiversity represents the variety of life on Earth, from animals and plants to fungi and single-cell organisms. It includes diversity across species, genetics, functions, evolutionary history and landscapes [1]. Biodiversity is crucial for the stable provision of ecosystem services such as clean air, water purification, food production and climate regulation. Local, regional and global biodiversity is therefore fundamental to human life and prosperity [2]. Despite conservation efforts, there is strong evidence that terrestrial biodiversity is declining, with average abundance of native species reduced by at least 20% since the early 1900s [3, 4]. Habitat degradation is an important driver of biodiversity loss, caused by land-use change and exploitation of natural resources; about 75% of the land surface is significantly altered by human activities [4]. Climate change, pollution and spread of invasive species are other major causes of decreasing biodiversity [5, 6].

Quantifying the biodiversity intactness of ecosystems, and associated drivers of decline, is a key concern for researchers. In response, a large number of databases, indices and models have been developed to measure the state of biodiversity in different ecosystems. They range from threat and extinction assessments [7], species- and community-level time series data [8], to statistical models linking biodiversity intactness to human pressures [9–11]. Such statistical models are used both for spatial extrapolation of biodiversity intactness on a global scale, and for temporal projections based on future socioeconomic development scenarios. These are important for tracking progress towards global and national biodiversity commitments,[1] and estimating the impact of policies and actions by governments and companies [12]. They are underpinned by field studies that have measured biodiversity around the world, which have been collated into large biodiversity databases like PREDICTS [13].

While the use of global-scale statistical models in biodiversity research has led to many important insights, there are some notable gaps in the literature: (i) There is a lack of systematic evaluation of model predictive performance [14, 15]. Evaluations often focus on model fit and statistical significance of parameters and not on out-of-sample predictive accuracy, which is crucial for models explicitly used for spatial and temporal extrapolation. Poor accuracy can have adverse implications on evaluation of biodiversity-related strategies and actions. (ii) The frequentist statistical models typically used, that rely on maximum likelihood estimation, are not well-suited to deal with sparse data, due to overfitting problems. This makes it difficult to derive insights about responses to human pressures on more granular biogeographic and taxonomic scales. (iii) Uncertainty in parameter estimates and outputs remain inadequately quantified [15]. Given how global biodiversity models shape our understanding of the state of ecosystems, and their impact on policy and decision-making, addressing these issues should be a high priority.

[1] For example, the UN Sustainable Development Goals, the Kunming-Montreal Global Biodiversity Framework, and the Global Convention on Biological Diversity

At the same time, Bayesian hierarchical models are increasingly used in population ecology. They provide flexibility in modeling complex ecological processes, are valid for small sample sizes and are able to quantify uncertainty well [16]. These models are well suited for modeling biodiversity data that contains complex hierarchies across biogeographic and taxonomic scales. Several smaller studies have used Bayesian models to estimate species abundance and occurrence based on environmental factors and human pressures. However, to the best of my knowledge, they have not been used for larger-scale biodiversity intactness modeling. The application of Bayesian hierarchical models in this setting is an opportunity to address the previously mentioned limitations.

The Biodiversity Intactness Index (BII) [10,11] is a well-recognized global-scale model adopted by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) as a core indicator for global biodiversity targets [17]. The model is designed to quantify the impact on biodiversity intactness from land-use and other human pressures. It is made up of two sub-models, one for species abundance and one for compositional similarity, from which final index scores are derived. It uses linear mixed effects models, trained on data from the PREDICTS database [13], to make spatial and temporal extrapolations. Despite its comprehensiveness, the BII is subject to the issues highlighted in (i)-(iii), particularly the lack of systematic evaluation. It therefore serves as a good case study and reference point for the use of Bayesian hierarchical models and a more systematic testing approach.

1.2 Aims, research questions and scope

This study aims to address some key aspects of the issues identified above. In light of issue (i), I implement the BII abundance model in [18] and systematically evaluate its predictive performance using cross-validation. This performance baseline serves as a reference for further model development. I then proceed with designing a Bayesian hierarchical model, utilizing a different hierarchical structure and additional covariates. The purpose of these changes is to improve the generalization of the model to unseen data. The performance of this model is benchmarked against the original BII model. The Bayesian hierarchical formulation should address issue (ii), by being able to estimate parameters on granular biogeographic levels without overfitting. This model should also give a better quantification of uncertainty than the frequentist counterpart, which indirectly addresses (iii) (however, an explicit analysis on this subject is not in scope). Finally, I compare the final index outputs of the two modeling approaches, to understand how they are impacted by differences in predictive performance. Based on the outlined steps, the specific research questions that I seek to answer are:

1. What is the out-of-sample predictive performance of the BII total species abundance model? What are the key limitations of the model?
2. How does the predictive performance of the Bayesian hierarchical model compare to the BII? What proportion of improvements is driven by differences in model structure and differences in model covariates, respectively?
3. What are the differences in final index outputs between the models?

Based on the answers to the research questions, I analyze the implications for the use of the BII,

and propose further pathways for improving predictive biodiversity modeling. In this study the scope is restricted to the BII species abundance model. I expect that a lot of the findings and approaches will apply to the compositional similarity model as well. The geographical scope is limited to tropical and subtropical forest biomes, as in the latest major BII paper [18], to facilitate benchmarking of results. The study does not address spatial extrapolation of intactness to sites that are not part of the PREDICTS database; this extension would allow the model to serve as a global biodiversity indicator.

The report is organized as follows: Sections 2.1 and 2.2 give a short introduction to different aspects of biodiversity and an overview of existing biodiversity indices and models. The theory of linear mixed models, used by the BII authors and in many other ecological applications, is covered in Section 2.3. This is followed by a detailed review of Bayesian hierarchical models in Section 2.4, including a deep-dive on prior selection. The background section is concluded with an overview of relevant evaluation metrics in Section 2.5. The method section starts with an overview of data sources and processing in Section 3.1, followed by a detailed description of the modeling approaches in Section 3.2. Results are presented in Section 4, focused on the comparison between the BII baseline and Bayesian hierarchical models. The results and their implications are discussed in Section 5, including limitations and proposed future work. The report ends with conclusions in Section 6.

2 Background

This section covers biodiversity and statistical fundamentals that are essential for understanding the methods and results of the study. Section 2.1 provides some key concepts and metrics used in biodiversity research, followed by an overview of data sources, models and indices for current tracking of biodiversity in Section 2.2. This includes a deep dive on the BII in Section 2.2.1. Linear mixed effects models, used in the BII and for many other ecological models, are presented in Section 2.3. Bayesian hierarchical models, which share many similarities with the former, are covered in-depth in Section 2.4. The section concludes with an overview of evaluation metrics relevant for hierarchical regression models in Section 2.5.

2.1 Some basic biodiversity concepts

The Convention on Biological Diversity (CBD) defines biodiversity as "the variability among living organisms from all sources [...] this includes diversity within species, between species and of ecosystems" [19]. Biodiversity plays a fundamental role in ecosystems on local, regional and global scales. It contributes to important ecosystem functions such as primary production (mainly photosynthesis), biomass production, nutrient uptake and filtration, pollination and decomposition [20]. These translate into final ecosystem services and goods, including timber and crops, climate regulation, erosion control, water and air quality, and recreation [2,21,22]. Diverse ecosystems are also more resilient in the face of changing conditions and extreme events [23]. Resilience captures both the ability of an ecosystem to recover from perturbations (such as floods, wildfires and human pressures) and how much pressure it takes to transform it to a new (often worse) steady-state. While current, sometimes low, levels of diversity might be enough to sustain ecosystem services today, it might not be enough to ensure resilience against future large-scale changes [23].

To greatly simplify things, biodiversity can be considered in terms of *type* and *scale*. Common types include taxonomic, functional and genetic diversity. Taxonomic diversity relates to the occurrence and abundance of different species across taxonomic groups [20], which is the most common diversity type studied in the literature. The taxonomic hierarchy, with our own species as an example, is shown in Fig. 1. The term taxon refers to any unit in the biological taxonomy. Except for individual species, a given taxon include several taxa from the lower levels of the hierarchy. Frequently used measures include species richness, the number of different species in an area, and species abundance, the sum of individuals of one or more species. Mean species abundance, the ratio between abundance and richness, is another common metric. Functional diversity relates to the traits and roles of species and groups in an ecosystem [20]. Functional overlap at different ecological scales, and reinforcement across scales, is associated with more diverse ecosystem services and greater resilience [24, 25]. Functional diversity often translates into response diversity, where species with similar functions respond differently to pressures [26]. Genetic diversity goes beyond the distinct physical traits of individual species, capturing additional difference between and within species [27].

Alpha, beta and gamma diversity are key concepts related to the scale of biodiversity. Alpha diversity captures the richness and abundance across species in the community of a certain geo-

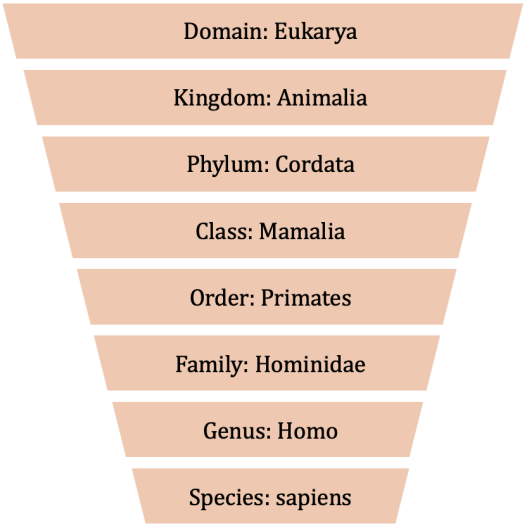


FIGURE 1: Illustration of the taxonomic hierarchy model used in ecology.

graphic area [28]. Beta diversity reflects the difference in species composition between two areas in a landscape [29]. Finally, gamma diversity combines alpha and beta to give an overall picture of biodiversity across larger geographical scales [29]. There are many competing approaches to calculating alpha, beta and gamma diversity. In the BII, the compositional similarity model quantifies (the inverse of) beta diversity [11]. The total species abundance model is on the scale of alpha diversity (measured for individual sites), but unlike traditional alpha diversity measures it does not consider the evenness of the abundance distribution at a site.

2.2 Measuring the state of biodiversity

Various indices and models have been developed to measure biodiversity, estimate past trends, and analyze future scenarios [3, 12]. This proliferation reflects both demand for accurate information to guide conservation policy, and the vast complexity of measuring biodiversity. There is a (paradoxical) desire for simple biodiversity metrics to track outcomes in a comparable way, yet also to have detailed information about certain species groups. Here I provide a non-exhaustive set of examples to illustrate different approaches, their advantages and drawbacks. I focus on terrestrial biodiversity, since marine and freshwater diversity are not in scope of the BII.

2.2.1 Global-scale spatial models

The BII, first proposed in 2005 and initially estimated using expert input [10], is a key example of a spatially explicit biodiversity model. The collation of the PREDICTS database [13] meant it could be quantitatively modeled on a global scale in 2015-16 [11, 30]. The BII definition and modeling approach developed at this time largely remains in place today. The taxonomic scope is quite broad compared to many other datasets [18], covering about 2% of all known species.[2]

[2]See the PREDICTS website for additional information.

The BII has been adopted by IPBES as a core indicator towards two of the CBD's global biodiversity targets (reducing risk of extinction and essential ecosystem services) [17]. IPBES is an international organization set up to strengthen the science-policy interface on topics related to biodiversity and ecosystem services, sometimes called the "IPCC for nature". It provides regular assessments of the state of nature, policy support and capacity building.[3] The BII is also included in the Essential Biodiversity Indicator data repository of the Group on Earth Observations Biodiversity Observation Network (GEO BON).[4] It has been proposed as a metric to measure functional biosphere integrity in the Planetary Boundaries framework, but was later replaced due to the not-so-well understood link between the BII and Earth system function [31]. The BII has previously been criticized for underestimating overall levels of biodiversity loss and showing limited agreement with other indicators [32]. Still, it remains commonly used in policy frameworks and in the scientific literature. It is one of few frameworks that provide a quantitative synthesis of the relationship between biodiversity and human pressures on a global scale.

The BII aims to estimate the relative biodiversity intactness of any given area, compared to a hypothetical pre-historic, pristine habitat, illustrated in Fig. 2. In the implementation developed by [11] that is used today, the BII consists of two sub-models, one for total species abundance, and one for compositional similarity [18]. The final score is a combination of the two sub-models, expressed as a percentage

$$\text{BII}_{0-100\%} = 100 * (\text{Total species abundance}_{0-1} * \text{Compositional similarity}_{0-1}) \tag{2.1}$$

where the subscripts indicate the intervals of the outputs. Sampling sites with minimally-used primary vegetation and absence of other human pressures are used as reference sites. The abundance and similarity models are trained on biodiversity and land-use data from PREDICTS. In addition to land-use type and intensity, population and road network density are used as covariates in the models. This implementation uses linear mixed models (covered in Section 2.3) where the model structure reflects how the PREDICTS data was collated: Random effects are fitted for individual studies and spatial blocks within studies.

One question mark already highlighted in the introduction is the generalization of the BII models to unseen data. In a recent paper [18], the BII is analyzed for tropical and subtropical biomes and all taxonomic groups. The abundance and similarity models exhibit reasonable levels of model fit, with conditional R^2 of 0.65 and 0.68, respectively. Conditional R^2 accounts for both population and group effects, in this case studies and blocks (see Section 3.3 for details on these metrics). However, the marginal R^2, which only uses information that would be available when predicting for new data (population effects), is only 0.05 and 0.12, respectively [33]. This indicates that the predictive capabilities of the models might be limited. This is not a BII-specific issue; essentially all global biodiversity models lack systematic evaluation of predictive performance [14, 15]. The focus has been on model fit and robustness of parameter estimates, with model selection typically based on significance tests and information criteria [34], not performance on out-of-sample data using cross-validation. This a concern given that predictions

[3] More information about IPBES can be found here.

[4] A description can be found here, including a link to the data repository.

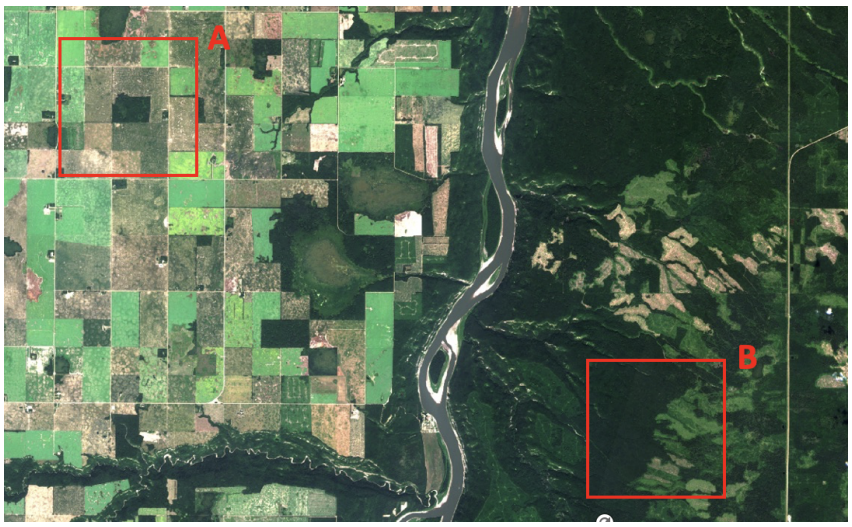


FIGURE 2: Example of a disturbed site (A) and a pristine site (B). Note that this is just an illustrative example and that B is not necessarily a completely undisturbed site. Image source: Canadian National Deforestation Monitoring System.

and scenario analysis from such models are used to guide policy and actions from public and private actors. This can lead sub-optimal planning and conservation practices.

GLOBIO is a similar well-known model, which been continuously developed and expanded since it was first launched in 2001. GLOBIO also targets the impact of human pressures on biodiversity, including climate change, land-use, roads, atmospheric nitrogen and hunting [9]. It calculates the mean species abundance of sites that are subject to a given pressure, and compares this against undisturbed baseline sites. This is then related to the extent of individual pressures in a set of regression models, one for each pressure type. The approach relies heavily on meta-study analysis of the impact of different factors, but also uses the PREDICTS database. One advantage of GLOBIO is the broad set of human pressures included, which go beyond those of the BII. However, there are some aspects of the modeling approach that can lead to bias in parameter estimates and predictions. First, GLOBIO uses separate regression models for each pressure type, restricting the data to cases where other factors are *considered* negligible or equal between sites [9]. This simplified approach, rather than trying to model the complexity and interaction of these processes together, could miss important relationships in the data. Second, the final model output seems to be at risk of under- or overestimation. If land-use is assumed to be the dominant factor impacting biodiversity, other factors are completely ignored (risk of overbias). Otherwise, the overall mean abundance is the product of the mean abundance of each pressure (risk of underbias). Similar to the BII, it has not been systematically evaluated in terms of predictive performance.

2.2.2 Other models and indicators

While the spatial models described above, especially the BII, are in focus of this study, there are other relevant biodiversity indicators to be aware of. The International Union for Conservation

of Nature (IUCN) Red List assesses the threat level to individual species, based on published studies and available global data [7]. The list covers about 7% of all known species and is widely reported. Reassessments are carried out occasionally, but not regularly across all taxons, so it cannot be readily used to track changes over time. The Living Planet Index (LPI) is another well-known biodiversity index. It uses time series data to track abundance of species and populations over time, with relative intactness measured against a baseline year. The LPI has been designated as an indicator towards several of the CBD's global biodiversity targets [8].

Other examples of species and community time series datasets include BioTIME [35] and the North American Breeding Bird Survey [36]. These datasets have been used to quantify overall trends in species abundance. A clear drawback is that they rarely include information about drivers of change, such as human pressures. They tend to focus on animals, especially well-known and iconic species, with less representation of other taxonomic groups. A recent paper indicates that the uncertainty of these trend estimates is much larger than previously thought, when properly accounting for all non-independence in the data [37].

Additionally, there is a group of indicators that can serve as proxies for biodiversity intactness. The Human Footprint (HFP) is a composite human pressure indicator that has been found to correlate with species range size, species richness, threat status and extinction risk [38, 39]. It captures land-use, population density, roads, railways and waterways, as well as night light presence. Another proxy for biodiversity, especially for plants and species that are highly dependent on vegetation, is biomass intactness [40]. A related metric is the Forest Landscape Integrity Index (FLII) that considers not just deforestation, but any infrastructure and activity that can lead to lower forest biodiversity, such as roads [41].

2.3 Linear mixed effects models

Ecological datasets are often structured in a hierarchical manner [42]. Figure 3 shows an illustrative example, where the population is divided into groups, which in turn consist of sub-groups. In the PREDICTS database, biodiversity measurements are taken at sites that are nested within spatial blocks, that are part of different studies [13]. In other settings, a set of sites can be sampled repeatedly over time to create longitudinal data. Data collected in this way violate the independence assumptions of most statistical and machine learning models. To deal with such non-independent data, linear mixed effects models (LMMs) is a common choice [43]. In addition to their widespread use in ecology and biodiversity research (such as the BII), understanding LMMs are also useful for understanding the Bayesian hierarchical models covered in Section 2.4, since they share many similarities.

A central concept in LMMs is the distinction between *fixed effects* and *random effects* [42]. Consider a standard linear regression model with a response variable y_i, which is a function of an intercept term β_0, an independent variable x_1, and a corresponding slope parameter β_1. We have the probability model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \mathcal{N}(\mathbf{x}_{ij}\boldsymbol{\beta}, \sigma^2) \tag{2.2}$$

where ϵ_i is independent and identically distributed (i.i.d) noise, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. The $\boldsymbol{\beta}$ parameters apply to the whole population of independent observations $\{x_i, y_i\}_{i=1}^N$. In other words, β_1

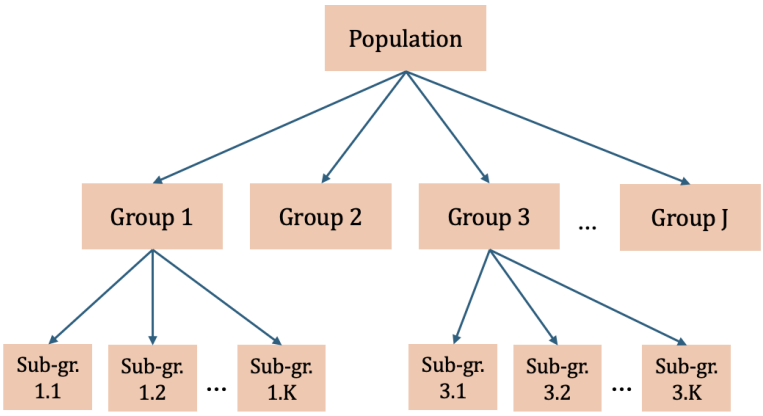


FIGURE 3: Illustrative hierarchical data structure with two levels. The population consists of J groups, and each group j consists of K sub-groups. Note that K can be different between groups j. This is nested structured, since each sub-group belongs to one and only one group.

represent the average effect across the population of a one unit change in x_i. We can view the population-level parameters as *fixed* unknown constants, to be estimated from the data.

In a hierarchical or grouped data structure, observations within a group are not independent of each other. Typically, we would expect that such observations are more similar to one another compared to observations from other groups (e.g. think of biodiversity measurements from two different ecoregions). Completely pooling these non-independent observations together can lead to underfitting and biased inference [42]. The other extreme of no pooling, fitting separate regression models for each group, tends to overfit the data, especially for groups with few observations [44]. It also implies that we are losing out on information that is interesting in its own right [42].

To accurately estimate the fixed effect parameter vector $\boldsymbol{\beta}$ in a grouped data setting, group-level random effects $\boldsymbol{\gamma}$ are included in the model, in addition to the population-level fixed effects [43]. If we let j represent different non-overlapping clusters or hierarchical groups in the data, and let i denote observations from a given group, then the example above becomes

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0j} + \gamma_{1j} z_{ij} + \epsilon_{ij} = \mathcal{N}(\mathbf{x}_{ij}\boldsymbol{\beta} + \mathbf{z}_{ij}\boldsymbol{\gamma_j}, \sigma^2) \tag{2.3}$$

In this case, the expectation for y_{ij} is a combination of fixed effect terms, β_0, β_1 and x_{ij}, and group-level random effect terms, γ_{0j}, γ_{1j} and z_{ij}. If we only included γ_0 this would be a random intercept model. The inclusion of z_{ij} and its slope parameter γ_{1j} makes it a random slope model [45]. This mixed effects model can flexibly capture the covariance structure induced by the grouping of the data, which includes variation between and within groups. Between-group variation is captured by the random effects, whereas within-group variation is represented by ϵ_{ij}. These errors are often assumed to be independent both within and between groups with fixed variance σ^2, such that $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ [43]. In essence, the inclusion of the appropriate random effects implies that the observations y_{ij} become *conditionally* independent [46].

There are several reasons why these effects are called *random*. One is that, in the case of observational studies, groups represent random samples from some population [43]. Another is that the group parameters represent random samples drawn from a population distribution of possible parameter values [42]. A third reason is that contrary to the fixed effects parameters, which are estimated directly via maximum likelihood, the random effects are sampled based on other quantities that are estimated from the data. The random parameters are *effects* because they shift the expected values of a group away from the population mean structure [43].

The parameters $\boldsymbol{\gamma}$ are either (i) independent draws from a common univariate distribution, or (ii) sampled from a shared covariance matrix [43], with (ii) being most common in software implementations. Let $\mathbf{y}_j$ denote the response vector for group j, $\mathbf{X}_j$ the fixed effects design matrix, and $\mathbf{Z}_j$ the random effects design matrix. These design matrices can fully or partially overlap, but do not necessarily need to. The general form of an LMM model is given by

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_j\boldsymbol{\gamma}_j + \boldsymbol{\epsilon}_j \tag{2.4}$$

In the case of (i), $\boldsymbol{\gamma}_j$ is a vector of independent values sampled from a univariate Gaussian:

$$\boldsymbol{\gamma}_j = \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_K \end{bmatrix}, \quad \gamma_k \sim \mathcal{N}(0, \sigma_\gamma^2), \ \forall\, k \in K \tag{2.5}$$

For (ii), $\boldsymbol{\gamma}_j$ is treated as a random vector sampled from a multivariate distribution defined by a covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\gamma}}$, such that

$$\boldsymbol{\gamma}_j \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\gamma}}) = \mathcal{N}\left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} \mathrm{Var}(\gamma_1) & \mathrm{Cov}(\gamma_1, \gamma_2) & \dots & \mathrm{Cov}(\gamma_1, \gamma_K) \\ \vdots & \vdots & & \vdots \\ \mathrm{Cov}(\gamma_K, \gamma_1) & \mathrm{Cov}(\gamma_K, \gamma_2) & \dots & \mathrm{Var}(\gamma_K) \end{bmatrix} \right) \tag{2.6}$$

The noise terms in both cases are assumed i.i.d, given by $\boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. The observed quantities in the model are $\mathbf{y}_j, \mathbf{X}_j$ and $\mathbf{Z}_j$. Maximum likelihood or restricted maximum likelihood is used to estimate $\boldsymbol{\beta}, \boldsymbol{\Sigma}_{\boldsymbol{\gamma}}$ and σ^2, while $\boldsymbol{\gamma}_j$ and $\boldsymbol{\epsilon}_j$ are sampled from these estimated quantities.

In cases with more than one grouping variable, there is a distinction between *nested* and *crossed* random effects [47]. In a nested structure, there are one-to-many relationships between groups at higher and lower levels. PREDICTS is a good example: Each spatial block has a unique identifier and only belongs to one unique study, and each study typically contains multiple blocks. Further, each site is nested within one and only one block. A crossed structure is essentially any structure that is not nested. For example, consider a case where sites are subject to a series of treatments (e.g. the sequential introduction of new species). Here, block and site would be crossed with treatment. A given model can include both nested and crossed effects.

There are some practical challenges of training LMMs compared to standard regression models. There needs to be enough groups in the data to achieve robust estimates of variance through the random intercept terms [48], and models can become unstable if the group sample sizes are highly unbalanced. It is also not clear how to gauge the statistical significance of the random effect terms. Another implementation challenge is choosing the specific random effects structure [42].

Random intercepts are primarily used to control for non-independence within groups, and thus common practice. Random slopes for continuous variables tend to give more accurate parameter values, but require significant amounts of data to fit accurately. While some researchers have suggested to fit a maximal random effect structure [45], this if often not feasible in practice due to data constraints [42].

Once the LMM has been trained, predictions for new observations depend on how much is known about them. Continuing with the PREDICTS example, consider a new site with covariate vectors $\mathbf{x}^*_{ij}$ and $\mathbf{z}^*_{ij}$, that we know belongs to an existing study and block. Then we can predict using all fixed and random effects, $\hat{y}^*_{ij} = \mathbf{x}^*_{ij}\boldsymbol{\beta} + \mathbf{z}^*_{ij}\boldsymbol{\gamma}_j$. However, if the group identity is not known, we are limited to using only the fixed effects for the prediction, $\hat{y}^*_{ij} = \mathbf{x}^*_{ij}\boldsymbol{\beta}$. The out-of-sample predictive performance will depend greatly on how much information can be captured by the fixed effect terms. This is a challenge that was highlighted for the BII in Section 2.2.

2.4 Bayesian hierarchical models

From a Bayesian perspective, the joint probability of the model parameters θ_j for the groups j, should reflect the dependency structures in the data. A hierarchical model structure can be achieved by assuming that the θ_j are drawn from a population distribution, parameterized by some *hyperparameter* ϕ. Since the hyperparameter ϕ is not known, we have to assign a prior distribution to it, a *hyperprior* $p(\phi)$ [49]. This induces a dependency structure into the model, where the prior distribution of the θ_j is dependent on the hyperprior. In a hierarchical Bayesian model, the joint prior distribution is

$$p(\theta, \phi) = p(\theta|\phi)\, p(\phi) \tag{2.7}$$

Consider a process that generates observations y_{ij}, where j denotes the group identity and i is an observation within a group, determined by group-level parameters θ_j. The complete Bayesian hierarchical model is given by

$$\begin{aligned} &\text{Likelihood: } p(y_{ij}|\theta_j, \phi) \\ &\text{Prior: } p(\theta_j|\phi) \\ &\text{Hyperprior: } p(\phi) \end{aligned} \tag{2.8}$$

Using Bayes' theorem we can derive the posterior

$$\begin{aligned} p(\theta_j, \phi|y_{ij}) &= \frac{p(y_j|\theta_j, \phi)\, p(\theta_j, \phi)}{p(y_{ij})} \\ &\propto p(y_{ij}|\theta_j, \phi)\, p(\theta_j, \phi) \\ &= p(y_{ij}|\theta_j, \phi)\, p(\theta_j|\phi)\, p(\phi) \\ &= p(y_{ij}|\theta_j)\, p(\theta_j|\phi)\, p(\phi) \end{aligned} \tag{2.9}$$

where the definition of conditional probability is used to factorize the joint probability $p(\theta_j, \phi)$. The simplification in the last step is because y_{ij} depends directly only on θ_j, not ϕ [49].

Bayesian hierarchical models are increasingly used in population ecology, offering a number of benefits: the posterior distribution is used for all inference, which provides full quantification of

uncertainty; inferences are valid for any sample size and does not rely on asymptotic assumptions (e.g. for rare species); informative priors can be leveraged in habitat types where data is sparse; missing data (common in field surveys) can be easily imputed; and they have the ability to work with intractable likelihood functions (e.g. non-Gaussian response variables with spatial dependence) [16].

For hierarchical models, the concept of exchangeability is fundamental [49]. A set of n observations y_i can be modelled as exchangeable if the joint probability $p(y_1, \ldots, y_n)$ is invariant to permutations of the indices. This is the underlying assumption made when we say that data is i.i.d. Applied to model parameters instead of observed data, it means that θ are independent samples from a population distribution and their joint distribution becomes

$$p(\theta) = \int \left(\prod_{j=1}^{J} p(\theta_j|\phi) \right) p(\phi) d\phi \tag{2.10}$$

As noted previously, hierarchical data is generally not i.i.d., so we need to think of exchangeability of observations within groups, or between entire groups, but not between observations that belong to different groups. Since the group identity j carries information, and the covariates x_{ij} also carry information, partial or conditional exchangeability has to be considered. If we appropriately model the group structure in a hierarchical model, and condition y_{ij} on x_{ij}, we still have an exchangeable model. That the model is assumed to be exchangeable does not mean that θ_j cannot differ between groups, just that we can assume that they are drawn from the same population distribution [49].

In Section 2.3 I introduced the concepts of complete and no pooling, as two extremes on the spectrum. One of the most powerful features of Bayesian hierarchical models is the effective use of partial pooling [50]. This means that inference accounts for information in each unique group, but also the information provided by all other groups, through the hierarchical structure of the model. The term shrinkage describes the way in which parameter estimates for a given group are shrunk (pulled towards) the mean of the supergroup to which it belongs. Using Fig. 3 as illustration, each group in the first level will be pulled towards the population mean, the sub-groups in group 1 will be pulled towards the mean of that group, and so on. If there is more uncertainty for a group, due to few observations or high within-group variance, the shrinkage will be stronger. Groups with less uncertainty will, on the other hand, be allowed to deviate from the mean. In a Bayesian hierarchical model, the shrinkage is very intuitive, since the groups are explicitly related through the hierarchical priors. The prior distribution of the group 1 parameter θ_1 is determined by the hyperprior ϕ, as in Eq. (2.8). In turn, θ_1 determines the priors for the sub-groups in group 1, such that $p(\omega_{k\in 1}, \theta_1) = p(\omega_{k\in 1}|\theta_1)\, p(\theta_1)$.

While Bayesian models are often associated with the use of (informative) priors, an equally important difference to frequentist models is that of expectations. The mean of the posterior distribution of θ is an expectation over all possible values, since θ is viewed as a random variable. This is fundamentally different from the frequentist perspective, which views θ as a fixed, unknown quantity. Here, θ represents a single value that maximizes the likelihood of the data. These two approaches can sometimes lead to quite different results.

2.4.1 Hierarchical linear regression models

Hierarchical regression models are powerful tools for complex regression problems with nested data. Consider a general model of data y_{ij} from J groups. The set of regression parameters $\boldsymbol{\beta}_j$ can be divided into two subsets. One subset, $\boldsymbol{\beta}_1$, is given independent priors and is implicitly modeled as exchangeable. The remaining parameters, $\boldsymbol{\beta}_2$, are assumed to be exchangeable and drawn from a population distribution governed by some mean and variance hyperparameters. These two subsets are essentially the fixed and random effects, $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}_j$ respectively, of the hierarchical regression model, similar to what we saw in Section 2.3 [49].

In a Bayesian setting, the terms fixed and random effects can be confusing since all parameters are viewed as random variables. We can instead think about the random effects as varying intercept and varying slope terms, and the fixed effects as population effects that do not vary by group. Consider a model where multiple parameters vary by group, with an assumed prior correlation structure between the group parameters. Let y_{ij} denote the outcomes for group j, let $\boldsymbol{\beta}$ be a vector of parameters that do not vary by group, and let α_j and $\boldsymbol{\gamma}_j$ be parameters that vary by group. The likelihood and group-varying prior for this model are given by

$$\begin{aligned} y_{ij} &\sim \mathcal{N}(\alpha_j + \boldsymbol{\beta}\mathbf{x}_{ij} + \boldsymbol{\gamma}_j\mathbf{z}_{ij}, \sigma_y^2) \\ \begin{bmatrix} \alpha \\ \boldsymbol{\gamma}_j \end{bmatrix} &\sim \mathcal{N}\left(\begin{bmatrix} \mu_\alpha \\ \mu_\gamma \end{bmatrix}, \boldsymbol{\Sigma}\right) \end{aligned} \tag{2.11}$$

where $\boldsymbol{\Sigma}$ denotes the covariance matrix of the group-varying parameters and where hyperpriors are assigned to the hyperparameters $\{\mu_\alpha, \mu_\gamma, \boldsymbol{\Sigma}\}$ [49]. The population effects $\boldsymbol{\beta}$ are given independent priors. We can note that this looks very familiar compared to Eqs. (2.3) and (2.6); indeed the LMM and hierarchical linear regression models are conceptually and structurally very similar.

There are many possible variations of this generic model. There can be more than one level of groupings of the data (as in Fig. 3). All parameters might vary by group, or just a subset. The data variance σ_y^2 can also be allowed to vary by group. The group intercepts α_j can be modelled as a function of covariates at the group level, for example $\alpha_j = \delta_0 + \delta_1 u_j + \epsilon_\alpha$. We can also add so called contextual effects, usually group averages of individual predictors, so that $\alpha_j = \delta_0 + \delta_1 u_j + \delta_2 \bar{x}_j + \epsilon_\alpha$. This can resolve problems with individual predictors x_{ij} being correlated with the group-level errors [44].

Making predictions for unseen observations y_{ij}^*, in case the group identity is known, amounts to sampling from the posterior distributions of $\alpha_j, \boldsymbol{\beta}$ and $\boldsymbol{\gamma}_j$ and using the sampled values to compute $\hat{y}_{ij}^*$. This is done for $s = 1, \ldots, S$ iterations to obtain a posterior predictive distribution. For new groups, one could use the posterior of the hyperparameters to sample group-varying parameters. However, this is only relevant when simulating what new groups are expected to look like. For prediction tasks, we have the same challenge as noted in Section 2.3, if the grouping variables cannot be inferred for new observations.

2.4.2 Choosing prior distributions

On a high level, the posterior distribution can be viewed as a weighted average of the prior and data likelihood. High certainty in the data due to high statistical power (e.g. large effect sizes,

large sample sizes, low noise) heavily updates the prior [51]. Conversely, a highly informative prior can exert strong influence over the posterior in cases of weak data [44]. Principally, the prior in Bayesian analysis is a way to encode knowledge about the problem analyzed, but in practice it is often used to stabilize sampling in complex models or treated as a nuisance to overcome in order to get the other benefits of Bayesian analysis [52]. Choosing prior distributions has attracted significant attention and debate in the Bayesian community.

Priors can be categorized based on the amount of knowledge they introduce into the model.[5] Non-informative priors typically refer to flat probability distributions, such as $\text{Uniform}(-10^3, 10^3)$ or very vague normal distributions, such as $\mathcal{N}(0, 10^5)$. Weakly informative priors might be somewhere on the scale of $\mathcal{N}(0, 1)$ to $\mathcal{N}(0, 10)$, but what is considered weak depends to a large extent on the specific data and problem analyzed [51]. Priors that are more specific than $\mathcal{N}(0, 1)$ are often viewed as informative. These examples assume that parameters are on a unit scale, such that the mean of a variable is zero, and one represents a standard deviation from the mean.

In practice, most Bayesian workflows are not based on a true prior belief about some process before data is collected or a model is defined. Instead, the prior distribution is required to perform the Bayesian analysis of data already collected. A prior will always be paired with a likelihood function, and is therefore dependent on the observed data; a prior that works well in one situation might not work in another [52]. While flat or vague priors can be non-informative in the sense of not encoding any prior knowledge, they are typically not recommended because they place a lot of probability density on unrealistic parameter values. Non-informative priors can lead to heavily skewed results for transformed response variables or lead to inference that approximates maximum likelihood estimates, negating the benefits of Bayesian modeling [51].

Weakly informative priors encode some structural information about the problem, without going all the way to encoding domain-specific information. They can therefore serve as default priors that achieve a certain level of regularization on parameter values [52]. The use of regularizing priors acknowledges that effect sizes in ecology, as in many other domains, are typically small. Large parameter estimates should only be supported by high statistical power [51]. This approach of course assumes that the weakly informative priors are truly weak, that is, appropriately defined in light of expected effect sizes or at least the scale or the problem. There should be some probability density on extreme but plausible values, but not on unrealistic ones. The prior predictive distribution is an effective method for validating priors before fitting the model. This involves generating samples of the outcome variable based on the prior distributions of all model parameters. This should be wider than the actual data distribution, but still on a similar order of magnitude [53].

For hierarchical regression models, there is often a large number of prior distributions that must be chosen. These include priors on the population parameters (that do not vary by group), priors for the group-varying parameters, as well as hyperpriors for these parameters, and priors on the data noise. The population-level regression parameters are assumed to be independent, and are often given zero-mean normal priors, $\mathcal{N}(0, \sigma^2)$, where σ should be set appropriately with regards to scale [51]. For a distribution with thicker tails that puts more density on more extreme values,

[5] The developers of the STAN package has an overview of up to date usage of priors in the Bayesian community, which is a great reference.

Student-T$(0, \lambda, \nu)$ can be used [54]. For $\nu = 1$ this becomes the Cauchy distribution and for $\nu \to \infty$ it approximates the normal distribution.

For the group-varying parameters $\boldsymbol{\gamma}_j$ it can be appropriate to assume some correlation structure, similar to what we saw for LMMs in Eqs. (2.6) and (2.11). One benefit is that between-group information about a certain parameter is more effectively used to estimate other parameters (i.e. more effective shrinkage). Another is that we explicitly acknowledge that the magnitude by which groups deviate from the population mean is correlated across intercept and slope parameters. However, this also increases the complexity of the model formulation and the sampling. It is possible to parameterize and scale the data in hierarchical models in a way that independent priors, which are typically more convenient, still make sense [55].

In the case of correlated group-varying effects, the parameters are assumed to be drawn from a multivariate normal $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The mean vector can be given a weakly informative or informative independent prior, like the regression parameters discussed above [56]. For $\boldsymbol{\Sigma}$, a logical prior choice would be the inverse-Wishart distribution, $\boldsymbol{\Sigma} \sim W^{-1}(\boldsymbol{\Psi}, \nu)$, since this is the conjugate prior of the covariance matrix [57]. In modern Markov Chain Monte Carlo (MCMC) samplers, however, it is more efficient to decompose the covariance prior into scaling factors and a correlation matrix, $\boldsymbol{\Sigma} = \mathrm{Diag}(\tau)\, \boldsymbol{\Omega}\, \mathrm{Diag}(\tau)$, where τ is a vector of scale factors [56]. The correlation matrix is given a so-called LKJ prior, $\boldsymbol{\Omega} \sim \mathrm{LKJ}(\eta)$. Small values of $\eta < 0$ assumes high correlations between parameters, whereas $\eta \to \infty$ implies uncorrelated parameters.

For scale parameters, such as the residual variance σ_y, covariance scale factors τ, or variance hyperparameters for regression parameters, σ_β and σ_γ, values are restricted to the positive line. Common choices like Uniform(a, b) and Inverse-Gamma(α, β) are actually not recommended, as they can result in improper posteriors and are generally less robust [49]. Better choices are distributions from the Half-Student-T family, of which the Half-Cauchy and Half-Normal are special cases. For robust regression, a Half-Student-T should be used for the residual variance [56].

2.4.3 Non-Gaussian likelihood functions

Many real-world processes generate non-normally distributed data. For example, ecological datasets often include binary observations (presence or absence), fractions, or counts of individuals [58]. The relationship between $\mathbf{x}$ and $\mathrm{E}(y|\mathbf{x})$ may also be non-linear. These cases do not preclude the use of linear models [49]. Generalized linear models (GLMs) extend the linear regression approach from normal data to other distributions [59]. This applies to hierarchical and mixed effects models for non-independent data as well, forming a class of generalized linear mixed effects models (GLMMs) [58].

In a GLM(M), the response variable y is assumed to follow a distribution from the exponential family. The generalization consists of three key components [46, 49]:

(i) First, there is a linear predictor component relating the design matrix $\mathbf{X}$ to the parameters $\boldsymbol{\beta}$, such that $\eta = \mathbf{X}\boldsymbol{\beta}$. This is identical to standard linear regression.

(ii) Second, we have a (non-linear) *link function* $g(\cdot)$, that relates the linear predictor η to the

observed variable. This is a one-to-one, monotonic and differentiable function such that $g^{-1}(\eta) = \mathrm{E}(y|\mathbf{x}) = \mu$. In fact, linear regression is a special case of the generalized linear model, using the identity link function, $g(\eta) = \mu$.

(iii) Third, there is a random component that specifies the distribution of y given the expectation $\mathrm{E}(y|\mathbf{x})$, such as σ in linear regression. This can involve a dispersion parameter ϕ [49].

The last part implies that the assumption of common variance for all observations, that $\sigma_i^2 = \sigma^2$ for all i (homoskedasticity), can be relaxed. Common non-Gaussian distributions in ecology where GLMMs can be applied include the Poisson (for count data), Binomial (for binary data), and Beta (for proportions or other values on the 0-1 scale). While GLMs offer flexibility to accurately model the data distribution, they can be more complex to implement and less computationally efficient. Transforming the response variable to be approximately normal allows the use of the linear models described previously, potentially simplifying the analysis. However, this can also run the risk of leading to biased inference [58].

2.5 Evaluation metrics for hierarchical models

Metrics for regression model evaluation can be organized into the broad groups of explained variance, information criteria and predictive performance. Several metrics from different groups typically need to be used in conjunction to provide a holistic view of model performance. In the first category, the coefficient of determination (R^2) is an ubiquitous metric for capturing the proportion of the variance in the data explained by the model [60]. For standard linear regression, this is typically defined in terms of residual and total sums of squares

$$R^2 = 1 - \frac{\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2}{\Sigma_{i=1}^{n}(y_i - \bar{y})^2} \tag{2.12}$$

where y_i are the observed responses, $\hat{y}_i$ are the predicted values, and $\bar{y}$ is the mean response value [61]. This is a natural definition for ordinary least squares regression, which aims to minimize the residuals $(y_i - \hat{y}_i)^2$. However, the definition is less straightforward for hierarchical models with mixed effects, which contain unexplained variance on several levels [42]. In this context, two complementary R^2 measures have been proposed: marginal R^2, the variance explained by the fixed effects only, and conditional R^2, the variance explained by the full model including fixed and random effects [34]. This method has gained significant support over recent years [42]. Instead of Eq. (2.12) we can view R^2 more directly in terms of variance explained by the model. For a model with fixed effects $\boldsymbol{\beta}$ and random effects $\boldsymbol{\gamma}_p$, with p representing different grouping levels (e.g. studies and blocks), the marginal R^2 is given by

$$R^2_{\mathrm{LMM(m)}} = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \Sigma_{p=1}^{P}\sigma_{\gamma_p}^2 + \sigma_\epsilon^2} \tag{2.13}$$

where $\sigma_\beta^2 = \mathrm{Var}(\hat{y}_{i\beta})$, that is, the variance of predicted values based on the fixed effect parameters. The $\sigma_{\gamma_p}^2$ terms represent the variance of predictions based on the random effects, and σ_ϵ^2 is the residual variance, $\mathrm{Var}(y_i - \hat{y}_i)$. The sum of the denominator terms is the total variance of

the observed data. The conditional R^2 is defined as

$$R^2_{\text{LMM(c)}} = \frac{\sigma^2_\beta + \Sigma_{p=1}^P \sigma^2_{\gamma_P}}{\sigma^2_\beta + \Sigma_{p=1}^P \sigma^2_{\gamma_p} + \sigma^2_\epsilon} \tag{2.14}$$

where the grouping factors have been added to the numerator. The quantities are easily calculated from the predictions and observed values. As with standard R^2, a value of one indicates a perfect fit. Additionally, this definition does place a lower bound at zero. One drawback is that it only provides information about the whole model, not the variance explained at each hierarchical level [34]. For a non-Gaussian model, the residual variance σ_ϵ is defined in a way that is specific to the error distribution and link function [34]. The approach has later been extended from random intercept to random slope models with non-Gaussian response variables [62–64].

In a Bayesian context, the traditional R^2 definition can also be problematic. First, it should ideally reflect the uncertainty in the model parameters θ, and second, with strong priors and weak data it is possible to obtain values greater than one [65]. Instead of point predictions $\hat{y}_i$, expected values conditioned on the unknown parameters are used: $y_i^{\text{pred}} = \text{E}(\tilde{y}_i|\mathbf{x}_i, \theta)$. Here, $\tilde{y}_i$ represents a future observation with covariate vector $\mathbf{x}_i$. Similarly to Eqs. (2.13) and (2.14) the numerator is the variance explained by the model, and the denominator consists of the variance explained and the residual variance

$$\text{Bayesian } R^2_s = \frac{\text{Var}_{\text{fit s}}}{\text{Var}_{\text{fit s}} + \text{Var}_{\text{res s}}} = \frac{\text{Var}(y_i^{\text{pred s}})}{\text{Var}(y_i^{\text{pred s}}) + \text{Var}_{\text{res s}}} \tag{2.15}$$

where $\text{Var}(y_i^{\text{pred s}}) = \text{Var}(\text{E}(\tilde{y}_i|\theta^s))$ is the variance of the vector of predictive means, and $\text{Var}_{\text{res s}} = \text{E}(\text{Var}(\tilde{y}_i - y_i^{\text{pred s}})|\theta^s)$ is the residual variance. The s indicate that we are using a set of posterior simulation draws $s = 1, \ldots, S$. For each draw, we can compute Eq. (2.15), and thus we get a distribution of R^2 values. The quantities can easily be calculated from the trained model. In the case of linear regression, y_i^{pred} is simply $\mathbf{x}_i\boldsymbol{\beta}$ and Var_{res} is the modeled variance of the likelihood function, σ^2. It should be noted that unlike the definition in Eq. (2.12), the denominator is not based on observed data and hence no longer fixed. Therefore, an increase cannot be directly interpreted as an improvement towards a fixed target [65].

Due to the previous absence of simple and general R^2 definitions for LMMs, information criteria have been commonly used for model comparison and selection in the ecological literature [34]. Information criteria capture the tradeoff between model fit (data likelihood) and complexity (number of parameters), so that complex random effects structures are penalized. The most commonly used definition include the Akaike information criterion (AIC) [66], Bayesian information criterion (BIC) [67] and Deviance information criterion (DIC) [68]. AIC and BIC are based on the log likelihood of the data under the parameters given by maximum likelihood estimation, with penalties on the number of parameters included. DIC is a generalization of AIC to hierarchical models, suitable for posteriors obtained via MCMC sampling, where the log likelihood is not readily available. More sophisticated measures for Bayesian models include the Widely applicable information criterion (WAIC) and pseudo leave-one-out cross-validation [69]. Information criteria can be used to compare two models trained on the same data, but it says nothing about whether a model is good in absolute terms. If back-testing using cross-validation

is very costly, they can serve as a proxy, but cross-validation is more robust and directly comparable between various models.

For predictive performance, there is a large family of measures based on the distance between actual and predicted values. Commonly used examples include the mean squared error (MSE) [70], and its root standardized form RMSE [71], the mean (and median) average error (MAE) [72], the mean percentage error (MAPE) [73], and the symmetric mean percentage error (SMAPE). One advantage of SMAPE is its bounds that facilitate interpretation, with a scale of 0-100% or 0-200% [74]. This makes it largely invariant to the scale of the data and expressed in a way that is interpretable and comparable across many settings. MSE and RMSE are very common in practice, but are not upper-bounded and do not have a natural unit of interpretation. MAE and MAPE are typically more natural to interpret, but suffer from the lack of an upper bound.

3 Methods

This section outlines the data and models used for the experiments. Section 3.1 provides an overview of the data sources used (Section 3.1.1) and key steps in the data processing (Section 3.1.2). The models are described in Section 3.2, starting with a detailed review of the abundance response variable in Section 3.2.1. The BII baseline model is briefly described in Section 3.2.2, followed by the Bayesian hierarchical model design in Section 3.2.3. A simple approach for calibrating biases in the models is presented in Section 3.2.4. The section ends with a summary of evaluation metrics used in Section 3.3 and brief implementation details in Section 3.4.

3.1 Data

3.1.1 Description of data sources

The PREDICTS biodiversity and land-use data, human population data and road network density data, mirror what is used the BII abundance model [18]. This is complemented by data on bioclimatic factors (from the same source used for the BII compositional similarity model) and data on topographic variables.

Biodiversity and land-use data: Projecting Responses of Ecological Diversity In Changing Terrestrial Systems (PREDICTS) is a major effort to collate biodiversity data from different studies (over 800 at this point, and increasing), into a structured database [13, 75]. The database contains abundance and occurrence data from 35,736 sampling locations in 101 countries, together with data on land-use and land-use intensity at those sites [76, 77]. There is a total of 4,318,808 records across 53,925 species, representing about 2% of all species named by science.[6] The studies in the database were conducted between 1984 and 2018. The database mainly covers animals and plants. The most common groups include arthropods (e.g. insects and other invertebrates), vascular plants and chordates (that includes sub-groups like vertebrate animals). Vertebrates are over-represented relative to their underlying diversity, while fungi and non-arthropod invertebrates are underrepresented.

The number of sites per study range from 2 to 754, with an average of about 44. Sites are in some cases organized into spatially adjacent blocks (255 studies), with on average about 4 sites per block. The sampling method is consistent for all sites within a study, but sampling effort can vary slightly [78]. Data from different sites within a study are thus comparable, but studies are not directly comparable. The number of taxons sampled per site varies greatly. The vast majority of diversity measurements are abundances, but these are expressed in different units depending on the study. The most common is individual counts, but proportions and density measures represent a significant share of all measurements. Land-use and land-use intensity for each site has been classified by the PREDICTS team, based on information in the original study. Land-use categories include primary vegetation, secondary vegetation (young, intermediate, mature, or indeterminate age), plantation forest, cropland, pasture and urban. Use intensity is classified as minimal, light or intense.

[6] Link to the PREDICTS website.

Human population data: Data on human population density (persons per 1 km) come from the Gridded Population of the World, v4.11 (GPWv4.11) dataset [79, 80]. The data has been collected from the 2010 round of Population and Housing Censuses, that took place between 2005 and 2014. The input data has been extrapolated to years 2000, 2005, 2010, 2015 and 2020 by the authors. The resulting raster data has a resolution of 30 arc-seconds, which is approximately 1 km at the equator. The census data has been adjusted to match the United Nation's World Population Prospects (UN WPP) country totals.

Road network data: Road network data are taken from the Global Roads Open Access Data Set, v1 (gROADS) [81]. It is a static layer of country road networks, joined at country borders and adjusted for topology. The data was collected between 1980 and 2010, with most of the data having no specific date information. The primary information is the length of road segments, which can be used to derive the road network density of an area.

Bioclimatic data: Bioclimatic variables come from the WorldClim v2.1 dataset [82]. These include annual trends, seasonal patterns and extremes for temperature and precipitation, and are designed to be meaningful in species distribution and other ecological models. The data represent averages over the years 1970-2000 and is available at 30 arc-second resolution.

Topographic data: The EarthEnv data repository contains, among other things, detailed topographic information at 1 and 10 km scales [83]. This includes variables for elevation, slope and terrain roughness and ruggedness. These are based on global elevation models derived from satellite imagery and LiDAR measurements.

3.1.2 Key data processing steps

To derive model variables from the datasets above, site coordinates from PREDICTS are buffered into square polygons of sizes 1, 10 and 50 km (using the `buffer` method from the shapely library). Before buffering, the coordinates are projected from global format to local UTM zones, to ensure that the resulting polygons are approximately equal in size, regardless of the distance from the equator. This is done using the `Transformer` class in the pyproj library. To be compatible with the global raster datasets on population density and environmental variables, the polygons are then re-projected into global format. I use the `zonal_stats` method in rasterstats to calculate average raster values of the pixels overlapped by each site polygon. Here I take the average of all pixels being touched by the polygon (as opposed to just the pixels whose center point falls within it).

For the road network data, I project the coordinates of the road line segments to local UTM zones and find the intersection between each site polygon and all nearby road segments, with `intersection` method in shapely. The total length of the intersecting road segments for each site is used as the road density metric. I interpolate the population density data between the available five-year increments, and extrapolate it back in time to the start of the PREDICTS data, assuming an exponential growth rate r, calculated as

$$r = \ln\left(\frac{y_{t_{\text{end}}}}{y_{t_{\text{start}}}}\right) / (t_{\text{end}} - t_{\text{start}}) \tag{3.1}$$

The road network, bioclimatic and topographic data are static layers, so no further processing is required. The PREDICTS data is filtered so that only abundance measurements are included. Entries where the land-use type or intensity is not given, are also filtered out, since these are key covariates in the model and must be available. To facilitate comparison with the most recent BII paper, the scope of the data used for modeling is limited to tropical and subtropical forest biomes [18].

3.2 Model specification

3.2.1 Total species abundance response variable

Let $\{\mathbf{s}_i\}_{i=1}^{G}$ be a set of sampling locations (sites) defined by (x, y) coordinates, sampled as part of a study $l = 1, 2, \ldots, L$ and in some cases assigned to a spatial block within the study, $b = 1, 2, \ldots, B$. At each site one or more taxons are surveyed. Let N_{ibl} be the true, unknown total species abundance at site i and let y_{ibl} represent the *observed* abundance. The observed abundance is obtained by counting all individuals of all species within the desired taxonomic scope. If there are $t = 1, \ldots, T$ species included in the survey and c_t represents the number of individuals of species t, the observed total species abundance is

$$y_{ibl} = \sum_{t=1}^{T} c_t \tag{3.2}$$

Observed abundance is a function of the unknown species abundance and the detection probability of the species surveyed, $y_{ibl} = f(N_{ibl}, d_{ibl})$. In order to explicitly model d_{ibl} we would need a set of covariates that are (at least partially) different from the ones that influence N_{ibl}, or some way of separating these models structurally. With the data described in Section 3.1.1, that is not feasible. In line with the BII [18], I therefore use the observed abundances y_i, scaled for differences in sampling effort within studies, as the response variable.[7] It seems reasonable that the human pressure covariates of interest mostly impact abundance, rather than detection. The analysis is restricted to all species at each site, rather than splitting the site-level abundance numbers by taxa. That represents a possible future extension of the model approach. As noted in Section 3.1.1, the observed y_{ibl} in PREDICTS are not always expressed as integer counts of individuals; they can also be fractions or densities. A Bayesian formulation does allow for combining data of different types in the same model, by specifying data-specific sub-models (e.g. using a negative-Binomial distribution for individual counts and a Beta distribution for fractions). However, this would add a significant amount of model complexity. I instead follow the BII approach of normalizing the data to unit scale, by dividing the raw measurements of each site by the maximum within each study [18]. This makes studies comparable regardless of the observed abundance measure used. It also reduces inter-study and inter-site variance; this can be computationally beneficial since individual abundance numbers can vary widely. The distribution of the resulting response variable is shown in Fig. 4, which is clearly right-skewed. This is very often the case with ecological data.

[7] Where measurements are reported in an effort-sensitive way, they are divided by the within-study sampling effort.

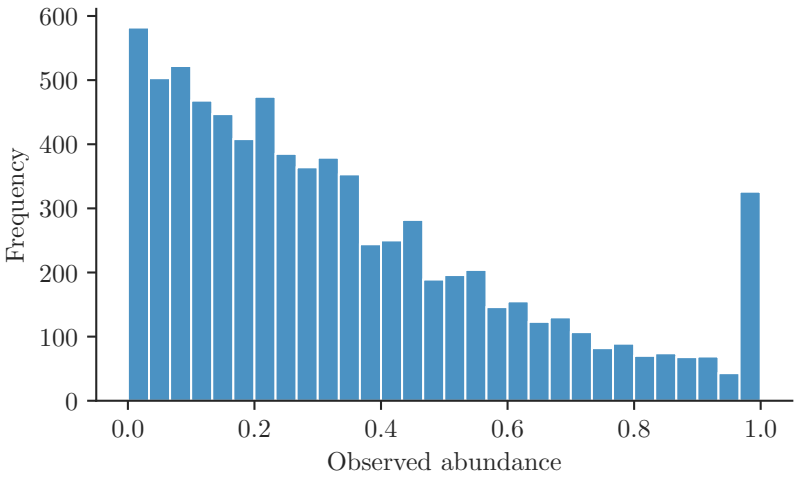


FIGURE 4: Distribution of observed abundance scaled to the 0-1 interval, by dividing each observation with the highest abundance site within each study.

While this approach has several advantages, the max-scaling leads to an inflated number of ones. This is because every study will have a maximum abundance site, and the scaled value for this site is always one. They represent about 3.2% of all sites in this scope (2.5% of the total PREDICTS data). Similarly, there is an inflation of zeros amounting to 2.8% of all observations (4.3% for the total data). Zero-inflation is common in ecological data. In this case it can represent either true absence of species or that species actually present are not detected during sampling. For individual taxons at a given site, true absence is not at all unlikely. At the same time, I would expect true zeros to be extremely rare if, hypothetically, all species at a given site were sampled. The other drawback is that the reference point within each study can be a site with any land-use conditions, not just minimally used primary vegetation sites. In fact, it is quite common for other types of sites to be the most abundant (see Fig. 14 in Appendix A). This is somewhat inconsistent with the aim of expressing the BII as a relative intactness measure. The reason appears practical, since only a minority of studies include a minimally used primary vegetation site, which would limit the data scope quite a lot.

3.2.2 BII baseline model

To answer research question (1) about BII predictive performance, I implement the total species abundance model as described in [18, 78]. For a site i, nested within a study l and block b, the normalized observed abundance is y_{ibl}. Let $\mathbf{x}_i$ be the fixed effects covariate vector, and let $\mathbf{z}_{il}$ and $\mathbf{z}_{ib}$ denote the covariates for the study and block levels, respectively. The response variable is square-root transformed, which makes the right-skewed data more symmetric. The likelihood is assumed to be Gaussian, which gives us the model

$$\sqrt{y_{ibl}} = \mathbf{x}_i\boldsymbol{\beta} + \mathbf{z}_{il}\boldsymbol{\gamma}_l + \mathbf{z}_{ib}\boldsymbol{\gamma}_b + \epsilon_i = \mathcal{N}(\mathbf{x}_i\boldsymbol{\beta} + \mathbf{z}_{il}\boldsymbol{\gamma}_l + \mathbf{z}_{ib}\boldsymbol{\gamma}_b, \sigma^2) \tag{3.3}$$

The fixed effect covariates $\mathbf{x}_i$ are identical to the original BII model in [18] and include categorical variables combining land-use type and land-use intensity (LUI); continuous variables of human population density at the 1 km scale ($\ln(1+x)$ transformed) and road network density at 1 and 50 km scales (cube-root transformed); as well as interaction variables between LUI and the continuous variables. Additionally, the mean population density within each study is used as a control variable (to account for bias in site accessibility between different studies). For certain land-use types, the intensity categories are grouped together: lightly and intensely used cropland, lightly and intensely used pasture, and all urban categories. All continuous variables are standardized such that $\tilde{x} = (x - \mu_x)/\sigma_x$. It should be noted that the normal distribution has support on $(-\infty, \infty)$ whereas the transformed response variable falls within the interval $[0, 1]$. This simplification is not explicitly explained or motivated in [18], but likely chosen because of computational convenience.

The study random effects, $\boldsymbol{\gamma}_l$, are aimed to capture differences in environmental and spatial factors, as well as sampling method and taxonomic scope, between studies. The block random effects $\boldsymbol{\gamma}_b$ should capture within-study differences in environmental factors. Since the final random effects structure is not reported in [18, 33], I experiment with three different structures:

1. The first version contains random intercepts for each study.
2. The second one adds intercepts at the block level. For studies that do not contain a block structure, the block id is equal to the study id, so this has no additional effect.
3. I then add certain random slopes at the study level, where $\mathbf{z}_{il}$ is a subset of $\mathbf{x}_i$, comprised of the categorical LUI variables.[8]

In the random slope model, the parameters $\boldsymbol{\gamma}_l$ are assumed to be correlated, following a multivariate normal distribution with mean zero, $\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\gamma)$.

3.2.3 Bayesian hierarchical model

The hierarchical structure in the BII model reflects how data was collected (studies and blocks). This is appropriate if the main purpose is explanatory: identifying which covariates are significant and what their effect sizes are. Study and block intercepts account for both biogeographic and sampling methodology differences, making observations conditionally independent. For predicting normalized abundance levels the structure is less ideal, since information encapsulated in the study and block parameters are not usable for making predictions on out-of-sample data. To overcome this problem, I shift to a hierarchical structure based on globally available information, illustrated in Fig. 5. This model will provide the input to research question (2).

The top level consists of biomes, reflecting large-scale differences in environment and underlying biodiversity patterns [84, 85]. Realms are not naturally a subset of biomes [86], but here I use them to subdivide the biomes into geographic areas that have distinct evolutionary paths that impact species composition. Finally, the ecoregions represent smaller-scale areas that capture differences in local landscape and environment [87]. The Bayesian formulation allows for a highly parameterized model without overfitting, through effective use of priors and shrinkage. That

[8] This structure was eventually confirmed by one of the BII main authors via email.

means I can estimate the full set of parameters at the most granular hierarchical level. Some realms and ecoregions contain relatively few observations, and their parameters will be pulled towards the means of the group to which they belong. For sub-groups where there is enough statistical power, the parameters is conversely able to pull away from those means.

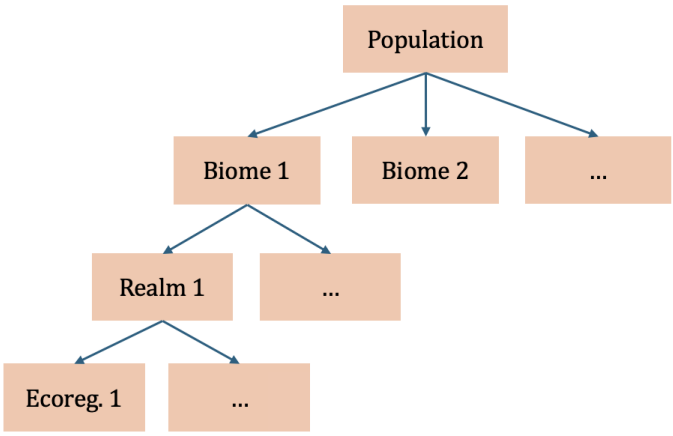


FIGURE 5: Refined hierarchical model structure based on established biogeographic groups and sub-groups.

I expect that the new hierarchical structure will capture a lot of biogeographic patterns previously absorbed by the study and block effects. This model will likely have a lower explained variance compared to the BII, since the hierarchical structure is less granular. On the other hand, it should be substantially better at making out-of-sample predictions. In this tradeoff between explanatory and predictive power, there is a fine-grained detail to be noted. By leaving out the study intercepts, there is no explicit control variable for differences in sampling methodology. Most of these difference should already be taken care of by the normalization procedure described in Section 3.2.1, but this could possibly induce a small bias in the parameter estimates. Put differently, the new hierarchical structure does not necessarily achieve the same level of conditional independence of the observations.

Formally, the hierarchical model structure consists of three levels with connected prior distributions. Let $b = 1, \ldots, B$ denote biomes, let $r = 1, \ldots, R$ denote biome-realm combinations (referred to as realms for simplicity) and let $e = 1, \ldots, E$ denote ecoregions. Realms are treated as subsets of biomes, $r \subset b$, and ecoregions are subsets of realms, $e \subset r$. The regression parameters are assumed to follow normal distributions, and variance parameters follow Half-Normal distributions. At the highest level, I define hyperpriors for the population-level regression parameters

$$\begin{aligned} &\text{Hyperpriors, } \alpha: \ \mu_\alpha \sim \mathcal{N}(a, b^2), \ \sigma_\alpha \sim \text{Half-Normal}(b^2) \\ &\text{Hyperpriors, } \beta: \ \mu_\beta \sim \mathcal{N}(0, b^2), \ \sigma_\beta \sim \text{Half-Normal}(b^2) \end{aligned} \tag{3.4}$$

where the choice of the scalar values a, b are discussed further down. Based on the hyperpriors, I sample a set of parameters for each of the three biomes. Since this is a quite complex model,

I start by assuming prior independence of the parameters at the biome level

$$\begin{gathered}\text{Biome priors: } \alpha_b \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2),\ \beta_b^{(k)} \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2) \\ \text{for } b = 1, \ldots, B,\ k = 1, \ldots, K\end{gathered} \tag{3.5}$$

where the k denotes the index of the regression parameters. This is more computationally efficient than assuming a correlated structure and using an LKJ prior (as discussed in Section 2.4.2). The downside is that the shrinkage will be less effective if there is in fact a highly correlated structure. This assumption can be tested by evaluating the posterior distributions [55]. To get the priors at the realm level, I first assume for the variance that $\sigma_{\alpha_r}, \sigma_{\beta_r} \sim \text{Half-Normal}(b^2)$. The distribution means for each realm are given by the biome priors to which they belong, which results in

$$\begin{gathered}\text{Realm priors: } \alpha_r \sim \mathcal{N}(\alpha_b, \sigma_{\alpha_r}^2),\ \beta_r^{(k)} \sim \mathcal{N}(\beta_b^{(k)}, \sigma_{\beta_r}^2) \\ \text{for } r = 1, \ldots, R \in b,\ b = 1, \ldots, B \text{ and } k = 1, \ldots, K\end{gathered} \tag{3.6}$$

The ecoregion priors are derived from the realm priors in an identical fashion:

$$\begin{gathered}\text{Ecoregion priors: } \alpha_e \sim \mathcal{N}(\alpha_r, \sigma_{\alpha_e}^2),\ \beta_e^{(k)} \sim \mathcal{N}(\beta_r^{(k)}, \sigma_{\beta_e}^2) \\ \text{for } e = 1, \ldots, E \in r,\ r = 1, \ldots, R \text{ and } k = 1, \ldots, K\end{gathered} \tag{3.7}$$

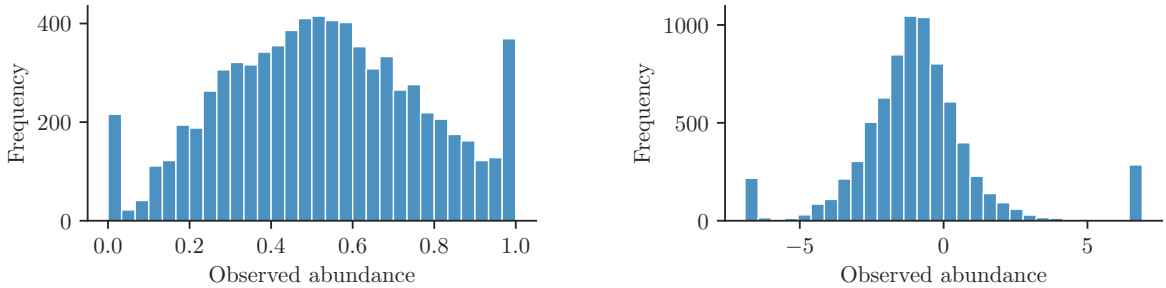
To improve sampling efficiency of the hierarchical model and avoid divergences, I use a non-centered parameterization of the group priors [56]. For instance, instead of sampling $\alpha_b \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$ directly, this is re-written as $\alpha_b = \mu_\alpha + \mathcal{N}(0,1)\sigma_\alpha^2$ in the model code.

A zero-one inflated Beta model appears to provide the best match to the distribution of observed abundances in Fig. 4. This is a finite mixture model consisting of three components with probabilities $p(y_i = 0), p(y_i = 1)$ and $p(y_i^*)$, where $y_i^* \sim \text{Beta}(\alpha, \beta)$ [88]. However, as previously noted, true zero abundances are extremely unlikely if the end goal is to estimate overall biodiversity intactness, and the inflated ones are just a scaling artefact. It therefore seems counterproductive to explicitly model this. The Beta distribution for the continuous part of the data is still relevant, as it has support on $(0, 1)$ and can flexibly fit skewed data. Nevertheless, using a Beta likelihood with a link function is less computationally efficient than a linear model. Further, it turned out to be very challenging to implement this for such a complex hierarchical model. For these pragmatic reasons I choose to work with a Gaussian likelihood function.

As an alternative to the square-root transformation of the response variable used in Section 3.2.2, I consider a logit transformation, which maps numbers on the interval $(0, 1)$ to the real line $(-\infty, \infty)$. In principle, this is more theoretically justified, since it matches the support of the Gaussian distribution. In practice, because the data needs to be bounded at two points close to 0 and 1 to perform the calculation, the logit transformed distribution also becomes bounded. The two alternatives are illustrated in Fig. 6. A final option is to use the untransformed variable (see Fig. 4), which is more interpretable in terms of estimated parameter values than the other options. It is not obvious upfront which approach is better, so I start by comparing them and continue to refine the model using the best-performing alternative.

For an observation in biome b, realm $r \in b$ and ecoregion $e \in r$, the likelihood becomes

$$t(y_{ibre}) = \alpha_e + \mathbf{x_i}\beta_e + \epsilon_i = \mathcal{N}(\alpha_e + \mathbf{x_i}\beta_e, \sigma_y^2) \tag{3.8}$$



(A) Square-root transformed.

(B) Logit transformed.

FIGURE 6: Distribution of response variable under different transformations.

where the noise $\epsilon_i \sim \mathcal{N}(0, \sigma_y^2)$ is given a common prior across all groups, $\sigma_y \sim \text{Half-Normal}(c^2)$. The function $t(\cdot)$ represents one of the transformations discussed above. It is somewhat challenging to choose the scalars a, b and c, since a Gaussian likelihood is used to model outcomes that are tightly bounded. By inspecting the prior predictive distribution, I find that $a = 0.5$ and $b, c = 0.1$ seem appropriate for the untransformed and square-root models, and $a = 0.5, b = 0.1, c = 0.2$ for the logit model. These values can appear highly informative, but in the light of the scale of the data, they are rather weakly informative: the prior predictive distribution is several times wider than the data distribution, but on the same order of magnitude. I evaluate the model (with the best transformation chosen based on the initial test) using four different sets of covariates:

1. The same covariates as in the BII baseline models.
2. Bioclimatic and topographic variables are added to the BII covariates. They include annual mean temperature and precipitation (1 km scale), temperature and precipitation seasonality (1 km), elevation (10 km) and terrain roughness (1 km).
3. More detailed LUI combinations, where light and intense use is no longer grouped together for cropland and pasture.
4. Simplified land-use variables, where only the type is considered (all intensities are bundled together).

I expect the variables added in (2) to explain additional variation in abundance patterns between areas with different environment and geography. These covariates are not of direct interest in terms of understanding how human activities impact biodiversity intactness. They still, however, offer more explanatory power to the model as opposed to letting these effects be absorbed by the intercept terms. The specific variables and scales are chosen based on having the highest correlations with the response variable. All covariates are evaluated for collinearity within their respective group, which reveals no correlation coefficients above 0.5. Finally, for (2)-(4) I drop the mean population density control variable, since it was both small and insignificant in [33].

3.2.4 A simple calibration model

With complex regression problems such as the ones above, there is a risk that a linear model will result in some bias of the predictions, violating the basic desired property of calibration: A calibrated model implies that, on average, predictions are equal to observed values, such that $E(y|\hat{y}) = \hat{y}$ [89]. We can also consider calibration for subsets of the data, for instance by calculating the calibration error for each decile of observed values. It might be that the model is calibrated overall, but is overbiased for small values and vice versa. Miscalibration of a linear regression model may result from missing variables, variables that have not been appropriately transformed using a non-linear function, or more complex non-linearity that the model is not able to fit.

Ideally, we can improve the base model to eliminate any bias. In practice, this is often infeasible, and calibration is still required. To calibrate the predictions, we need to find a function such that $g(\hat{y}) = \hat{y}^*$, where $\hat{y}^*$ are adequately calibrated [89]. One simple approach is to fit a polynomial model to the residuals, where the observed values are used as predictors. Let $r_i = y_i - \hat{y}_i$ denote the residuals from one of the Bayesian hierarchical base models above. To test the impact of calibration in a simple fashion, I train a standard (non-Bayesian) linear regression model with the following specification:

$$r_i = \beta_0 + \beta_1\sqrt{y_i} + \beta_2 y_i + \beta_3 y_i^2 + \epsilon_i \tag{3.9}$$

The polynomial terms should correct for any systematic non-linearity in the relationship between predictions and observations. The intercept will shift the regression line if needed to correct an overall bias. This model is designed to be simple and general enough to not overfit the training data, so that the out-of-sample predictions also improve. The calibrated predictions are obtained by subtracting the calibration model predictions from the original predictions, $\hat{y}_i^* = \hat{y}_i - \hat{r}_i$.

3.3 Model evaluation approach

To evaluate the models, I use a combination of explained variance (R^2), predictive performance metrics (mean and median absolute error) and standard model checks. Information criteria are not included since they have limited value once out-of-sample testing using cross-validation is performed. For the BII baseline model, I leverage the definitions of conditional and marginal R^2 for LMMs described in Section 2.5 [34]. For the Bayesian models, I generally follow the definition of Bayesian R^2 in [65] but make one adaptation for consistency with the LMM R^2 calculations: Instead of using the posterior mean of σ as the residual variance, I calculate it from the predictions, $\mathrm{Var}(y_i - \hat{y}_i)$. In addition to consistency, a benefit of this approach is that it works regardless of the likelihood function or response variable transformation, without doing any extra calculations or adjustments. Since R^2 is a measure of model fit, it is based on predictions made on training data, where predictions and observed values are *not* back-transformed to the original 0-1 scale.

For the Bayesian models, I draw 1,000 samples from the posterior predictive distribution to obtain distributions of R^2 (and the error metrics). The values reported are the posterior means of these distributions. Note also that for the Bayesian models, there is no difference between

conditional and marginal R^2, since all group information used for training can also be used for out-of-sample data.

Testing and interpreting the predictive performance of ecological models on out-of-sample data, by using back-testing, is not trivial. In most regression problems there is some level of measurement error that can impact the representativeness of back-testing results, but this is on a different order of magnitude when dealing with species occurrence or abundance data. Detection probabilities are sometimes fairly low, with many repeated samplings required to approximate the true abundance of a site, so the observed values become very noisy approximations of the true state. This might explain why this type of evaluation is largely lacking in the literature [14, 15]. Still, doing back-testing on unseen data will give an important indication of the model's performance, as long as we use representative sampling and observation errors are not systematic. This is especially true if the back-testing results are used to compare the performance of several models. In principle, this approach is not different to ground-truthing predictions by collecting new observations, as suggested in [18].

For predictive performance, I use a combination of mean and median absolute errors. Among the family of distance metrics from Section 2.5, these are the most appropriate given the boundedness of the data. Errors can be naturally interpreted as percentage point deviations on a 0-100% scale, which corresponds to the scale of the final BII output. Moreover, errors are proportional along the range of observed values between zero and one. The problem with SMAPE for this particular problem is that errors of the same absolute size appear larger for small observations than large ones. In addition to calculating the mean and median absolute errors across all data points, I also calculate them for the top and bottom quartiles of observed values, to reveal potential bias for subsets of the data. For some initial model comparisons among the Bayesian models they are calculated on training data, since cross-validation is very costly for complex Bayesian models. For the final benchmark results, the metrics are based on 5-fold cross-validation, taking the mean and standard deviation across test and train folds. The folds are constructed using stratified sampling, to ensure presence of all biome-realm-ecoregion combinations when testing the Bayesian models. To enable this, I drop all groups with less than ten observations (which also helps with sampling speed). Predictions and observed values are always back-transformed to the original scale for these performance metrics. After applying these filters on top of the ones described in Section 3.1.2, there are 7,591 observations (sites) used for the modeling.

I also perform a set of standard model checks on each model after training. Calibration scatter plots show the relationship between observed and predicted values, which ideally should be approximately linear. Plotting residuals against observed values effectively reveals any bias in the model predictions, or if the assumption of constant variance, that σ^2 is a fixed parameter, is violated [43]. Given the assumption of a Gaussian likelihood for all models, the residuals should be approximately normal. For the Bayesian models, the posterior predictive distribution should also approximate the observed data distribution. Convergence in the MCMC sampling is checked by computing $\hat{R}$ values across all sampling chains: The stationary distributions of all the chains should converge, regardless of the initial values. The number of iterations should also yield sufficiently high effective sample size (ESS) numbers, and there should be no divergences in the chains.

An important aspect in this particular setting is that the model predictions $\hat{y}_i$ are not the final outputs from the abundance model. To obtain the final outputs that contribute to the index score in Eq. (2.1), each prediction is divided by the model prediction for a reference site, which we recall is a minimally used primary vegetation site with zero population and road density [11,18]. Mathematically, this is equivalent to dividing every prediction by the model intercept. This implies that an improvement in terms of mean and median absolute errors does not translate one-to-one into an improvement of the final outputs. To approximate this impact, I also analyze the distributions of the final outputs of the BII baseline and Bayesian hierarchical models. For the BII baseline, the outputs are based on a global set of fixed effects parameters and a global intercept; for the hierarchical model, the parameters and intercept are estimated on the ecoregion level, $e = 1, \ldots, E$. Let $\hat{y}_i^{(\text{out})}$ denote the final outputs for the $i = 1, \ldots, G$ sampling sites in scope, such that

$$\begin{aligned} \text{BII baseline: } \hat{y}_i^{(\text{out})} &= \frac{\hat{y}_i}{\beta_0} \text{ for } i = 1, \ldots, G \\ \text{Hierarchical: } \hat{y}_{ie}^{(\text{out})} &= \frac{\hat{y}_{ie}}{\alpha_e} \text{ for } i \in e,\ e = 1, \ldots, E \end{aligned} \tag{3.10}$$

I also compare the distribution of the model predictions directly, $\hat{y}_i$ and $\hat{y}_{ie}$.

3.4 Practical implementation details

The BII model is implemented in pymer4, which provides a Python interface to the commonly used lme4 package for LMMs in R [90]. The Bayesian hierarchical models are implemented using the PyMC library [91], one of the leading Bayesian Python libraries alongside Stan [56]. The default sampler of both PyMC and Stan is the No-U-Turn Sampler (NUTS), which is a Hamiltonian Monte Carlo sampler with adaptive path length [92]. This is widely considered to be the best MCMC sampler today for most problems. To speed up the sampling, I use the numpyro NUTS backend powered by the numerical computing library JAX [93–95]. For most of the experiments, I use 3,000 tuning iterations and 1,000 draws across four parallel sampling chains. For the complexity of the model, this is enough to obtain accurate estimates of the posterior means. The target acceptance rate is set to 0.99, which is a quite high value, to avoid divergences when sampling the complex posterior distribution. All experiments are run on an Apple M2 CPU. I attempted to use local GPU acceleration, but Apple's jax-metal extension was not compatible with PyMC.

4 Results

This section contains a detailed performance evaluation of the models described in the previous section, starting with a summary view in Section 4.1. The next few sections dive deeper into these results, first the BII baseline model in Section 4.2 and then the hierarchical base model results in Section 4.3, including the calibration model. The impact on the final biodiversity intactness outputs are analyzed in Section 4.4.

4.1 High-level summary

The Bayesian hierarchical approach is clearly outperforming the BII baseline model in terms of predicting normalized total species abundance, as seen in Fig. 7. This shows the best-performing Bayesian hierarchical model and the calibrated version of that model. The hierarchical base model has a $(0.24 - 0.16)/0.24 = 33\%$ lower mean absolute error rate, and a 52% lower median absolute error rate. The improvements are driven by the shift to the biome-realm-ecoregion hierarchical structure and additional variables on bioclimatic conditions, topography and more granular land-use information. The higher R^2 of the BII baseline model is conversely explained by the study-block structure, which provides a more granular fit to the training data. Calibration of the base model leads to a substantial further improvement of predictive performance, representing a 58% and 68% improvement over the BII baseline, respectively. That calibration works so well on out-of-sample data indicates that there is a persistent bias in the base model across all training data samples.

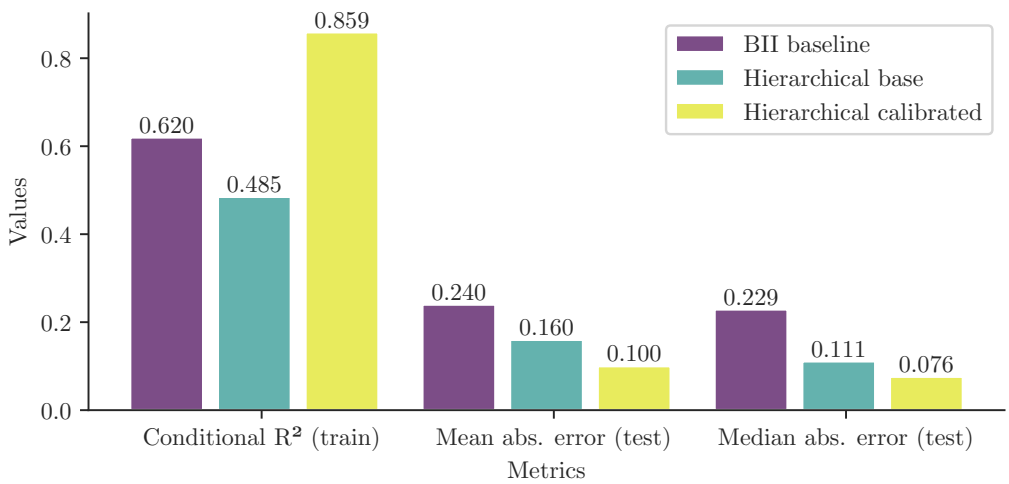
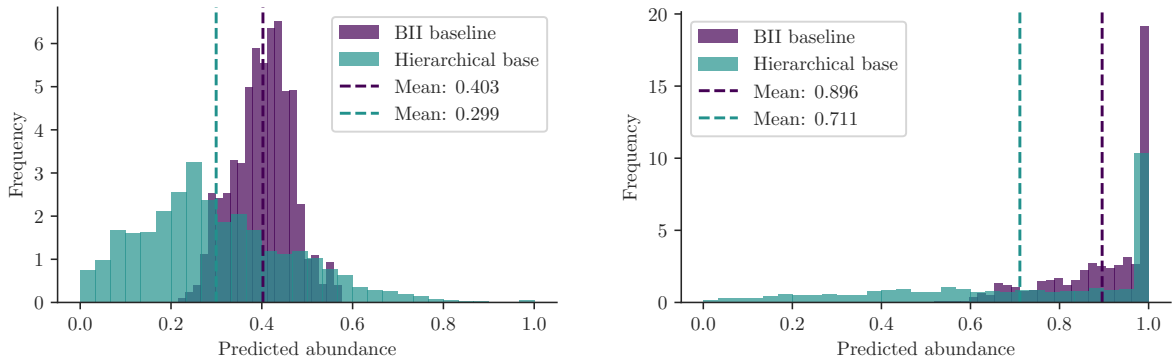


FIGURE 7: Performance metrics for the BII baseline model, the Bayesian hierarchical model with the complete set of land-use combinations, and the calibrated version of that model. Conditional R^2 is calculated based on predictions using all group-varying parameters, on the training set. Predicted and observed values are not back-transformed before R^2 calculation (except for the calibrated model, where this was a step in the calibration process). Mean and median absolute errors are based on out-of-sample test data, with values back-transformed to the original 0-1 scale.

As previously discussed, it is unclear a priori how these predictive performance improvements translate into differences in the final outputs. The results of this analysis are displayed in Fig. 8. The distributions of predicted abundance values in Fig. 8a align quite well with the calibration plots of the respective models shown in Section 4.2 and 4.3. The BII baseline model predictions are tightly clustered around the mean, since predictions are only made using the fixed effects. The hierarchical model predictions look more reasonable, considering the data distribution in Fig. 4. Issues arise when attempting to scale the predictions using the reference predictions, as evident by Fig. 8b. There is a significant amount of predictions that become larger than one, which need to be capped. The underlying reasons for this pattern are further analyzed in Section 4.4.



(A) Distributions of predictions from the hierarchical base and BII baseline models

(B) Distributions after dividing predictions by the relevant reference predictions.

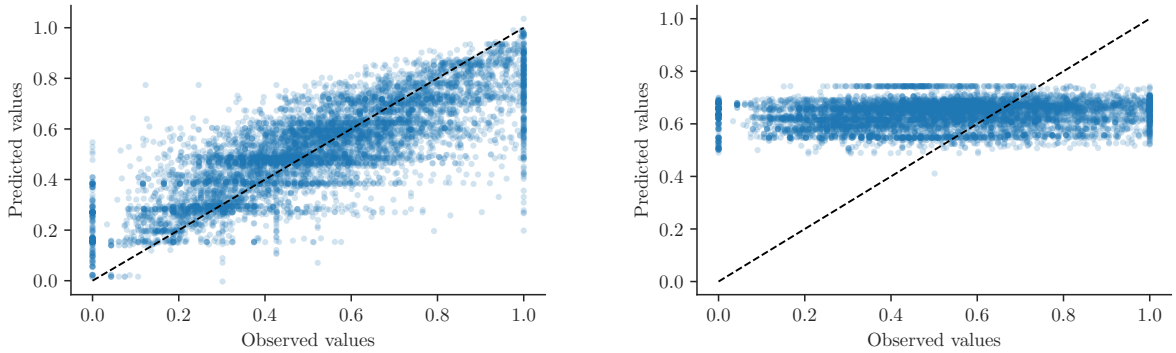
FIGURE 8: Predictions from the BII baseline and hierarchical base model, before and after dividing by the reference predictions.

4.2 BII baseline model deep dive

The conditional R^2 of the BII abundance models amount to 0.486 for the study intercept model, 0.539 for the block intercept model, and 0.620 for the study slope model. The corresponding marginal R^2 is 0.053, 0.048 and 0.047. These are roughly in line with the results reported in [18, 33]. Comparing conditional and marginal R^2, it is clear that the random effects explain the vast majority of the variation in the data. The largest relative increase in explained variance comes from the inclusion of the study intercepts. Varying intercepts at the block level only contribute slightly to the conditional R^2, likely because sites within a given study are already quite similar and because far from all studies contain block information. There is a somewhat larger shift in explained variance when including the study random slopes; given the large geographical and environmental spread of the data, it seems reasonable that there are meaningful differences in the slope parameters between studies.

The large discrepancy between the marginal and conditional R^2 is clearly visualized in Fig. 9 (for the study slope model; note that values have not been back-transformed for these calibration

plots). In the conditional plot on the left, there is a clear, albeit very noisy, relationship between predictions and the corresponding observed values. In the marginal plot, there is virtually no such relationship at all. Almost all predictions end up in a band of values between 0.5 and 0.8, centered around the intercept, regardless of the observed abundance. The model never makes predictions that are close to zero or one. This can be further understood by analyzing the magnitude of the model parameters (see Table 5 in Appendix A). While there are some parameters that are statistically significant, most of them are quite small in terms of effect size. It is also interesting to note that when moving from the random intercept models to the random slope model, the number of statistically significant fixed effects drop from 17 to 4.



(A) Conditional predictions using full random effects structure.

(B) Marginal predictions based only on the fixed effects.

FIGURE 9: Conditional (A) and marginal (B) predicted values (y-axis) compared to observed values (x-axis) for LMM with study slopes and block intercepts. Predictions and observed values are on the square-root scale (not back-transformed).

Out-of-sample test results are shown in Table 1, with mean values and standard deviations calculated over the five cross-validation folds. Predictions are made using only the fixed effects structure, and all values have been back-transformed to the original scale. The overall mean and median absolute errors do not necessarily look so concerning compared to what is indicated by the model fit in Fig. 11b, which highlights the limitation of working with aggregate metrics. Looking at the top and bottom quartiles of the observed data, we see that the errors are almost twice as large for the bottom quartile as the average. The results are very consistent across all three model specifications. The back-transformed predictions of the study slope model compared against observed values are shown in Fig. 10. This looks almost identical to Fig. 11b, except that the values have been shifted down.

Table 1: BII baseline out-of-sample performance

	Study intercept	Block intercept	Study slope
Mean abs. error (test)	**0.236** (0.004)	**0.236** (0.004)	**0.240** (0.003)
- Top/bottom quartile	0.194/0.409	0.198/0.409	0.198/0.416
Mean abs. error (train)	**0.235** (0.001)	**0.236** (0.001)	**0.239** (0.002)
Median abs. error (test)	**0.224** (0.003)	**0.222** (0.004)	**0.229** (0.005)
- Top/bottom quartile	0.184/0.391	0.187/0.390	0.190/0.394
Median abs. error (train)	**0.223** (0.002)	**0.223** (0.002)	**0.230** (0.002)

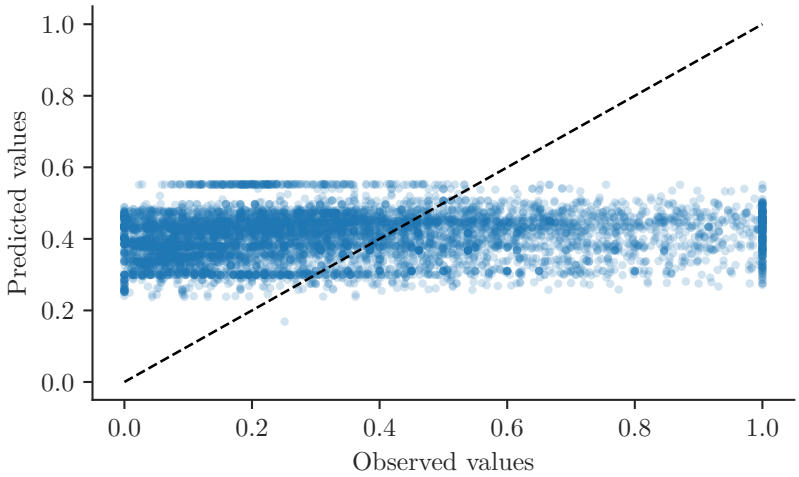


Figure 10: Predictions from the BII baseline model, back-transformed to the original 0-1 scale.

4.3 Bayesian hierarchical model deep dive

The first experiment is a comparison between models with untransformed, square-root and logit response variables. The square-root model has the highest R^2 of 0.456, compared to 0.416 of the untransformed model and 0.367 of the logit model. In-sample predictions are also more accurate for this model: mean absolute errors of 0.146 (square-root), 0.15 (untransformed), 0.153 (logit); median absolute errors of 0.105, 0.116 and 0.111, respectively. The only exception is that the square-root model has somewhat bigger errors for the top quartile of observations. The residuals of the square-root model are normally distributed and symmetric, while the other models have more skewed residual distributions (before back-transformation). A similar picture is given when inspecting the posterior predictive distribution, which approximates the square-root transformed observed data very well. The difference to the untransformed model is not surprising, since the square-root makes the skewed data more symmetric and approximately normal. The logit posterior is much wider than the observed data, despite the narrow priors. This is likely because the inflated zeros and ones are transformed into values quite far from the

mean, which pulls the tails of the posterior towards larger absolute values. The assumption of prior independence between regression parameters is tested by computing pair-wise correlations between all parameters at population and biome-realm level. All correlation coefficients are between -0.2 and 0.2, with the vast majority between -0.1 and 0.1.

Based on the results of the first experiment, I proceed with the square-root model with independent regression parameter priors for the rest of the experiments. Results of testing different sets of covariates are reported in Table 2, with error metrics calculate on in-sample data (not based on cross-validation due to computational time). The addition of bioclimatic and topographic variables provides a marginal improvement over using only the BII covariates. The small magnitude of this increase is a sign that a lot of environmental attributes are already accounted for in the new hierarchical structure. Including the full set of LUI combinations, without any aggregations, does not substantially impact the model fit or in-sample predictions. This can either imply that there are no major differences between lightly and intensely used pasture and cropland, or that there is not enough signal in the data for the model to properly identify such patterns. Interestingly, simplifying the land-use attributes to not account for intensity at all, results in a clearly worse model fit. Comparing this to the BII covariate model, the loss from this simplification is greater than the gain from the bioclimatic and topographic factors.

Table 2: Hierarchical model in-sample performance by covariate selection

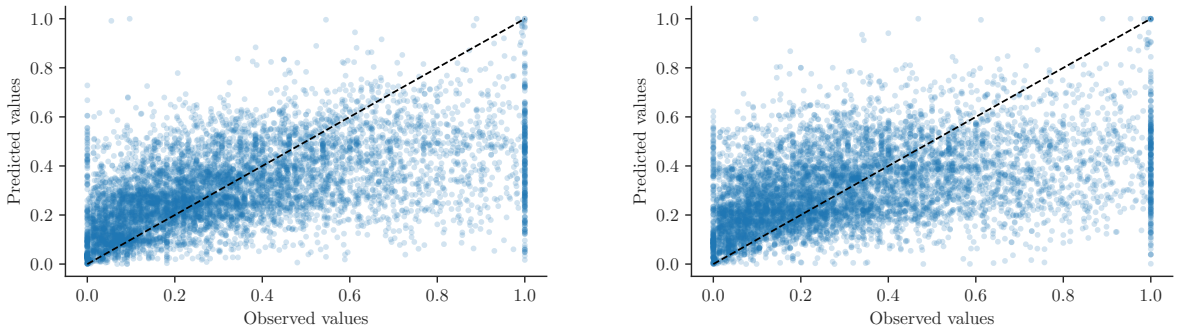
	BII covariates	Added bio/topo	Granular LUI	Simplified LUI
Conditional R^2	**0.456**	**0.483**	**0.485**	**0.434**
Mean abs. error	**0.146**	**0.140**	**0.140**	**0.148**
- Top/bottom quartile	0.280/0.104	0.266/0.095	0.266/0.095	0.283/0.107
Median abs. error	**0.105**	**0.099**	**0.099**	**0.106**
- Top/bottom quartile	0.252/0.089	0.238/0.077	0.238/0.077	0.259/0.093

An interesting question is how the most complex model, with the granular LUI combinations, and the simplified LUI model, generalize to unseen data. Table 3 shows the out-of-sample predictive performance of the two models. The differences are quite small, but the model with more granular land-use covariates delivers better performance also on new data. This suggests that despite the higher complexity, this model is not overfitting to the training data. The differences between results on test and training data do not indicate significant overfitting either. The results for both models are significantly better than the BII baseline model in Table 1, which we also saw in the summary in Section 4.1.

The calibration plots of the two models are shown in Fig. 11. They are fairly similar, exhibiting the same underbias for large observed values. The distribution of the predictions is somewhat tighter for the granular LUI model, which explains the better predictive performance.

Table 3: Hierarchical model out-of-sample predictive performance

	Granular LUI	Simple LUI
Mean abs. error (test)	**0.160** (0.008)	**0.177** (0.019)
- Top/bottom quartile	0.300/0.103	0.317/0.111
Mean abs. error (train)	**0.138** (0.001)	**0.146** (0.001)
Median abs. error (test)	**0.111** (0.008)	**0.132** (0.018)
- Top / bottom quartile	0.274/0.082	0.273/0.095
Median abs. error (train)	**0.097** (0.002)	**0.105** (0.001)



(A) Model with most granular land-use and intensity variables (Granular LUI).

(B) Model with only land-use categories, with intensities bundled together (Simplified LUI).

Figure 11: Calibration plots of the hierarchical models with full set of LUI covariates (A) and simplified LUI covariates (B). Predictions and observed values are back-transformed to the original 0-1 scale.

4.3.1 Impact from model calibration

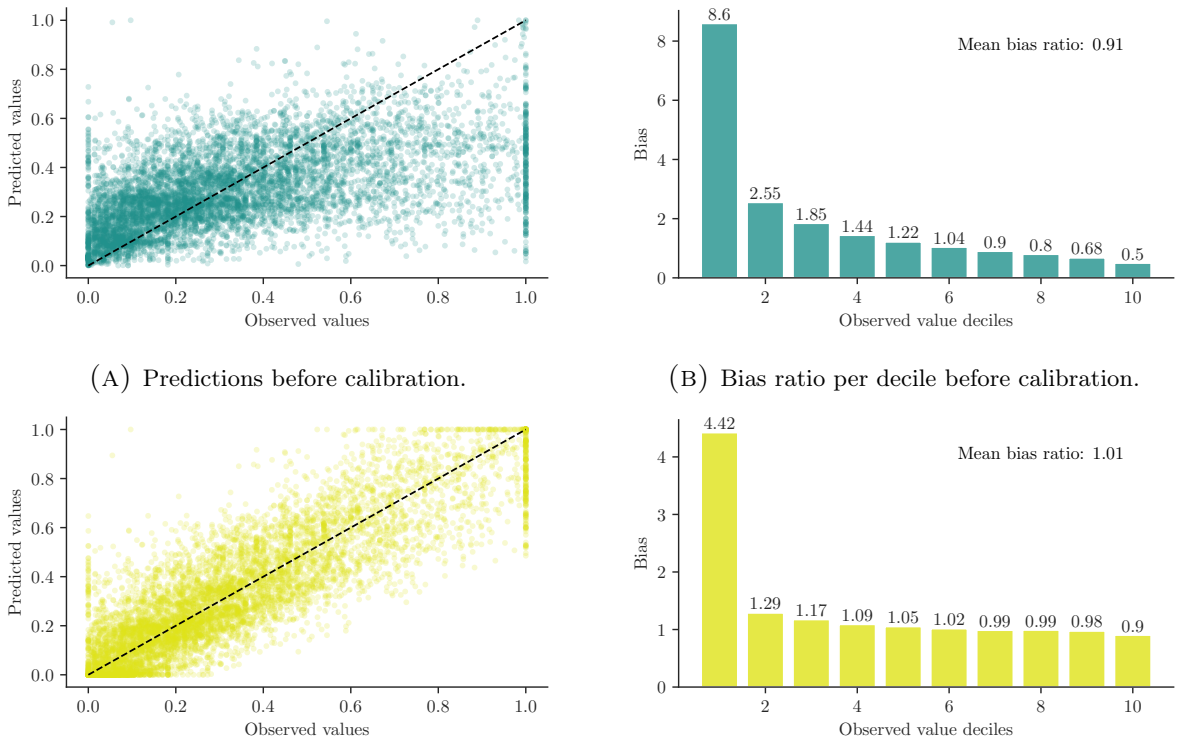
There are two calibration issues at play in the model results above. First, back-transforming predictions and observed values by squaring them induces an overall bias relative to how the models were trained, since the back-transformation (and the transformation itself) is not linear. Second, regardless of transformation, the models make overbiased predictions for small observed values, and the opposite for large observations. This problem is most obvious with the marginal predictions from the original BII model (Fig. 11b), which are heavily underbiased for large observed values. Even the conditional BII model, with granular block intercepts, is subject to some bias (Fig. 11a). The hierarchical models exhibit more bias at both ends of the spectrum compared to the conditional BII model.

It turns out that the simple calibration approach outlined in Section 3.2.4 is very effective at correcting these systematic biases, as can be seen in Table 1. The results show that this approach actually generalizes well to out-of-sample data. The calibration not only improves the overall results, but the difference between test and training data errors are not larger than pre-calibration.

Table 4: Out-of-sample predictive performance with calibration

	Granular LUI		Simple LUI	
	Base	Calibrated	Base	Calibrated
Mean abs. error (test)	0.160	**0.100**	0.177	**0.106**
- Top/bottom quartile	0.300/0.103	0.128/0.078	0.317/0.111	0.117/0.064
Mean abs. error (train)	0.138	**0.084**	0.146	**0.085**
Median abs. error (test)	0.111	**0.076**	0.132	**0.081**
- Top/bottom quartile	0.274/0.082	0.112/0.055	0.273/0.095	0.096/0.044
Median abs. error (train)	0.097	**0.084**	0.105	**0.067**

To get a closer understanding of what the calibration is doing, we can look at Fig. 12. Pre-calibration, the prediction-observed ratio, $\Sigma\hat{y}_i/\Sigma y_i$, is around 0.9 which implies the model is underbiased overall. There is also a clear trend going from overbias to underbias when the observed values are split into deciles. Post-calibration, while still noisy, there is now a clearer linear relationship between predictions and actual values, and the systematic bias is largely gone.



(A) Predictions before calibration.

(B) Bias ratio per decile before calibration.

(C) Predictions after calibration.

(D) Bias ratio per decile after calibration.

Figure 12: Results of model calibration of the granular LUI model. The bias plots show the mean of predictions divided by the mean of observed values. Deciles are based on the observed values. All values have been back-transformed to the original scale.

4.4 Distribution of model predictions and outputs

The results above show that the Bayesian hierarchical models are significantly better than the BII baseline model at predicting normalized species abundance. We also saw in Fig. 8 that the BII approach of dividing every prediction by the model prediction for a reference site, to obtain a relative intactness measure from the normalized abundance predictions, leads to unexpected results. Since the reference prediction equals the model intercept, which is around 0.44 after back-transformation for the BII baseline model, this results in a large upward shift that pushes many values outside the 0-1 range, as seen in Fig. 13. There is no mention of how this is handled in [18], but it seems reasonable that values cannot fall outside the 0-1 range, given the BII index definition. However, capping the distribution leads to a large spike at one and an overall mean value of 0.896, which seems very high. The mean of the hierarchical model final outputs (Granular LUI, before calibration) at 0.711 is more reasonable, but the distribution has a similar spike at one, seen in Fig. 8b. In this case, the intercept used to divide the model predictions is different for each ecoregion.

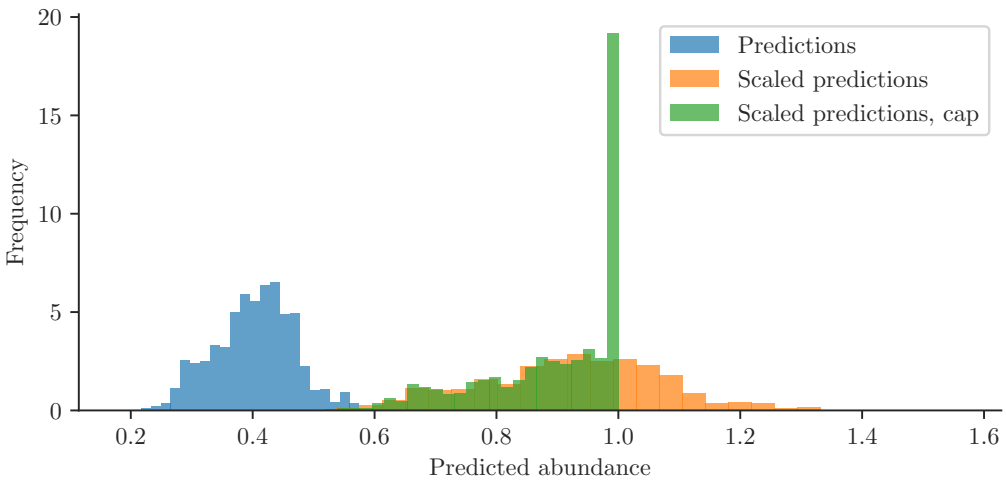


FIGURE 13: Distribution of BII baseline model predictions, outputs after scaling by the prediction for a reference site, and after capping the scaled outputs.

5 Discussion

This section contains an in-depth analysis of the results of the experiments, related to the research questions. I start by comparing and discussing the performance of the different models in Section 5.1, including the impact on the final outputs in Section 5.1.1. For simplicity, I only compare numbers from the BII model with study slopes, to the Bayesian hierarchical model with the granular LUI combinations and its calibrated version. This does not impact the generality of the discussion. Data and model limitations are covered in Section 5.2. The implications of the results and the use of the BII are discussed in Section 5.3, followed by some proposed avenues for future work in Section 5.4.

5.1 Model comparison and evaluation

The discrepancy between conditional and marginal R^2 and lack of predictive evaluation of the BII model in [18], motivated research question (1) about out-of-sample performance and model limitations. While the overall mean and median absolute errors (0.24 and 0.229, respectively) might not appear problematic on the surface, Fig. 10 clearly shows that there is very little relationship between predicted and observed normalized abundance values. This pattern is due to only a small fraction of the overall data variation being captured by the fixed effects, and that their effect sizes are quite small. If the BII had been designed mainly as an explanatory model, the study-block random effects structure would have been entirely appropriate. But in a prediction model it creates a large gap between in-sample and out-of-sample performance. In summary, the answer to (1) is that the BII abundance model does not perform well on out-of-sample data, driven by the structure of the model.

The Bayesian hierarchical model gives a substantial improvement of out-of-sample predictions: a mean absolute error of 0.16 vs 0.24, which is a 33% improvement; and a median absolute error of 0.111 vs 0.229, which represents a 52% improvement. The influence of the study-block random effects on the BII baseline conditional predictions indicates that environmental factors cause more of the variation in species abundance than human pressure factors, even when data is normalized within each study. This can explain why the structure of biomes, realms and ecoregions works so well in the Bayesian hierarchical model. This is also boosted by the new bioclimatic and topographic variables and more granular land-use covariates. That the Bayesian model is indeed differentiating parameters quite significantly between realms and ecoregions, can be seen in Figs. 15, 16 and 17 in Appendix A. The loss in explained variance, with R^2 of 0.485 compared to 0.620, is not surprising given that blocks provide a very granular fit to the data, and that studies also reflect differences in taxonomic scope and sampling methodology. This answers the first part of research question (2), showing that the predictive performance of the new model is clearly better.

For the second part of question (2), the influence of the new hierarchical structure seems larger than the influence of additional covariates, as suggested by Table 2. Even though these are in-sample metrics, and thus not fully comparable with the BII test results in Table 1, there is a large decrease in errors with the new structure while keeping the BII covariates fixed. Still, adding the bioclimatic and topographic variables does provide a higher explained variance. Similarly,

the difference between the model with the full set of land-use combinations and the simplest one (in Table 3) indicates that the human pressure variables also play an important role in model fit and predictions.

The Bayesian approach is a crucial enabler of the new hierarchical model structure. The final model is highly complex, with three hierarchical levels, 12 categorical variables, 10 continuous variables and 24 interaction terms, which results in 3,948 regression parameters in total. It is not tractable to fit this model using maximum likelihood and even a greatly simplified model would have a high risk of overfitting. Validity for small sample sizes and regularization through shrinkage and weakly informative priors implies that the Bayesian model handles the inherent complexity of the problem well [49]. The differences between train and test errors in Table 3 do not indicate any major overfitting to the training data, despite the complexity. However, there are also good arguments for favoring the simplified land-use model, since the performance differences on out-of-sample data in Table 3 are not that great. This is especially true in a production setting, where land-use type can quite easily be derived from real-time satellite data, while land-use intensity requires less dynamic data and more assumptions [11].

Even though the hierarchical base model represents a significant improvement, it still suffers from bias, especially underbiased predictions for large observed values. The calibration results in Table 4 show that it is possible to correct these biases using a very simple approach. My initial concern was that this would overfit to the training data, but the calibration leads to substantial improvements on out-of-sample data too. Of course, it would be even better to correct the bias by improving the base model. Possible changes include different or additional non-linear transformations of covariates, more interaction terms and adding more variables to the model, which are all candidates for future work. But given the complexity of the problem and the noise in the data, I do not believe that any of these would fully address the bias, so there is still a strong case for calibrating the model. The approach in this study should be regarded as a quick prototype; there are other approaches that should be tested in the future.

5.1.1 Impact on final index outputs

In many situations an improved predictive model directly translates to better insights and decisions. Unfortunately, it is not so straightforward in this setting, illustrated by the distribution of predictions compared to final outputs in Figs. 8 and 13. Recall from Section 3.2.1 that the abundance response variable is not explicitly defined in relation to a pre-historic reference condition, since normalization is done by dividing each observation by the maximum in each study. This gap between end goal and implementation explains the additional step of dividing every model prediction by the reference site prediction, to shift from normalized abundance to relative intactness outputs, used in [11, 18]. It is not clear from those papers how the outputs above the 0-1 interval are handled. I can see two options: Either they are capped before entering the overall BII formula in Eq. (2.1) (together with the similarity model output), or the outputs of that formula are capped. Regardless, the distribution of final outputs for both the BII baseline and the Bayesian hierarchical do not seem reasonable given the number of exact ones.

Only in the case when all regression parameters are negative would this approach produce outputs on the 0-1 interval. In that case, the intercept, which represents the abundance of a

reference site, would be the highest predicted value. Primary vegetation sites with low population and road density would get predictions close to one, and all other types of sites would have lower values. Conceptually, this aligns with the overall formulation of the BII, where disturbed or less intact sites are considered worse from a biodiversity perspective. In practice, some of the land-use categorical covariates and interaction terms between land-use and population and road density, have positive parameters that explain the predictions above the intercept in Fig. 10 (also seen in Table 5 in Appendix A and in [33]). This can be driven by accessibility and sampling bias, but also that the total species abundance metric is sensitive to outlier species. To overcome this problem, the response variable either needs to be redefined, or we need to constrain the model to negative parameter values.

Putting this issue aside, the Bayesian hierarchical model appears to be a better foundation for deriving final outputs. While the BII baseline model has one set of global parameters, the hierarchical model estimates parameters down to ecoregion level. It thereby accounts for how biodiversity responds to pressures in different areas, not just the relative level of pressures. The fact that the numerator of Eq. (3.10) is more accurate, which has been shown earlier, and that the denominator is more granular, implies an overall improvement in outputs. In an ideal world, these final outputs could be compared against observed values on the same scale. The problem is that not every study contains a baseline site that meets the criteria. And even if that had been the case, normalizing the observed data with just one or a few reference points and using them to measure accuracy would not be representative. Research question (3) about the impact on final outputs therefore remains partially unanswered.

5.2 Data and model limitations

Limitations of the PREDICTS data are discussed in [11, 13, 18, 30] and apply here as well. One gap is that it does not distinguish between native and invasive species, potentially contributing to the issues with outlier species impacting the response variable, discussed above. When land changes from primary vegetation to other types, it can lead to an influx of new species that have a high affinity for the new habitat type, causing an inflation of the abundance numbers for non-primary vegetation sites. Time since conversion to a new land-use type is also an important variable, since it takes time for a habitat to reach a new steady state. Since this information is only available for about 40% of observations, it cannot readily be used in the model. The results may be biased if many sites have not reached their new steady states post conversion. As with any large-scale collation of data from many independent studies, there might be classification issues. The reported land-use type is the predominant one at a site, and the data does not distinguish between cases where this makes up different proportions of the total area. There could also be inconsistencies in how land-use intensity has been classified between studies. Even though PREDICTS is taxonomically broad compared to many other datasets, it still only contains a small fraction of all known species, which means that results are only representative for animal and plant taxa.

Since the road network data is static and was collected between 1980 and 2010 [81], it might not correspond well to the presence of roads at the time of sampling, especially in remote or less developed areas. The population density data is only available in five-year increments, and

interpolation is uncertain, particularly in less populated areas. If either the start or end year is zero for a site, it is not possible to do a proper interpolation, so all values have to be set to zero. More dynamic data would result in better estimates of these parameters.

One model limitation is that the new hierarchical model structure does not explicitly control for differences in taxonomic scope and sampling methodology, as discussed already. The abundance data normalization should already handle most of this this, but there could be some remaining influence not accounted for. Another is the simplifying assumption of a Gaussian likelihood function, which occasionally (but very rarely) results in predictions outside the 0-1 range. More importantly, even with the new structure, predictions for ecoregions that have not been sampled before might be worse, since the model would have to fall back to the biome-realm parameters. The systematic bias in the hierarchical base models has been discussed at length already, and is a priority for future work on the model. Finally, compared to the frequentist LMM, the Bayesian models are more complex to implement and runtime is substantially longer.

5.3 Implications of the results on the BII

In the light of the findings in this study, there should be some caution in using the BII for policy and decision-making purposes. In addition to out-of-sample predictive accuracy and the output scaling approach that appears to skew results to one, there are some other issues that should be highlighted. The multiplicative formulation of the overall BII in Eq. (2.1) is not explicitly motivated. As an example, an area that is predicted to be 80% intact by two separate models would be considered only 64% intact overall. A more common and easily justifiable approach when combining outputs from several models is to weight them, where the weights sum to one. A more fundamental potential problem with this formula is that a total species abundance model and a compositional similarity model are conceptually different entities, where the outputs are not immediately comparable. Additionally, the total species abundance metric is sensitive to abundant outlier species, and thereby sometimes insensitive to increasing species richness, and does not reflect distribution evenness.

If understanding relative biodiversity intactness is the specific goal, it seems logical to mainly focus on beta diversity, comparing every disturbed site with one or more reference sites. This is what the BII compositional similarity model does [18]. This is not a perfect comparison, as even such reference sites are often subject to some human disturbance, but it removes the impact of highly abundant species and specialists that are present *because of* land-use change and other pressures. This focus makes sense if we are more concerned with diversity levels that support well-functioning and resilient ecosystems, rather than abundance or richness itself. An alpha diversity model can be a relevant complement to the beta diversity model, but forcing them together into one overall index number can make the results less transparent and understandable. Finally, it should be noted that the BII similarity model uses the same post-prediction scaling approach, and hence likely suffers from the issue with outputs larger than one.

5.4 Proposed future work

Given the continued decline of global biodiversity, and future growth scenarios putting even higher stress on ecosystems, there has never been a greater need for accurate and representative

global-scale biodiversity models. The findings in this study clearly show that there is large room for improvement in how such models are designed and implemented. Rapid technological development is opening up many new opportunities for better monitoring of ecosystems [96]. High-frequency satellite data, including LiDAR and multispectral imaging, enable real-time detection of changes in land-use and vegetation. Paired with good predictive models, it would give us tools to quantify the state of biodiversity on a continuous basis and highlight problematic hotspots as they arise.

There are several components from the BII, GLOBIO and other existing models that can be reused in an improved modeling approach. The Bayesian hierarchical modeling approach developed in this study is an important contribution, since it demonstrates the feasibility of modeling complex ecological processes on a global scale with relatively good out-of-sample performance. There are some general (non-exhaustive) design principles I believe should be applied when developing improved models: (i) The biodiversity metrics should be designed to answer the specific questions at hand, even if that means they cannot be combined into a single metric; (iv) the model should ideally combine a data-driven (machine learning) approach with a clear causal structure and domain knowledge about key ecological processes; (ii) it should be designed to facilitate systematic evaluation; (iii) the model should reflect differences in how biodiversity responds to human pressures across biogeographic and taxonomic scales; (v) it should be transparent and interpretable.

In addition to what has been discussed, a fundamental limitation today is the amount and quality of biodiversity data available, at least in consistent formats. One avenue for future research is to develop models that are able to combine data from several different sources. Occurrence data that lacks information on land-use can be augmented by extracting data from historical satellite images. The same holds for species time series data. Advancements in eDNA barcoding and remote sensing (drones, bioaccoustics) provide additional tools for efficient collection of biodiversity data. These approaches require less expertise than traditional taxonomic surveys and can therefore be leveraged at greater scale and at lower cost. Observational studies can be combined with eDNA and remote sensing input to create a more holistic picture of the biodiversity of habitats. This can lead to a better understanding of both underlying biodiversity patterns and the impact of human pressures.

There are also additional covariates that can be investigated. Measures like NDVI (normalized difference vegetation index), vegetation biomass intactness [40] and the Forest Landscape Integrity Index [41] are partially reflective of human pressures, and partially of underlying biogeography. All of them provide continuous gradients that can complement the categorical land-use variables in PREDICTS. The Human Footprint contains some complementary covariates on human infrastructure, namely railways, waterways and presence of night lights [38, 39]. On country level, institutional quality indicators can be used as proxies for environmental protection [97]. There are also direct measures of environment regulation, but they only cover a subset of countries [98]. It would also be relevant to include climate change impact and pollution.

6 Conclusions

Biodiversity is crucial for the provision of ecosystem services and goods that our societies rely on [2, 23]. There is scientific consensus that terrestrial biodiversity has declined significantly over the last century, driven by land-use change, built infrastructure, climate change and pollution [3–6]. Understanding how human activities impact biodiversity is important for guiding policy and action from the public and private sectors [12]. There is a great need for accurate and interpretable global models that relate biodiversity to human pressures. Given their reach and policy implications, it is important that such models are thoroughly evaluated in terms of relevance, performance and robustness. This has long been a gap in the literature [14, 15]; well-known models such as the BII and GLOBIO have not been systematically tested for predictive accuracy. Other issues identified in the literature include inadequate quantification of uncertainty and lack of differentiated response estimates across biogeographic and taxonomic scales. These findings motivated the aim of this study, which was to evaluate the existing BII implementation and investigate how Bayesian hierarchical models could be leveraged to improve large-scale biodiversity modeling. The research questions and the associated results are summarized below.

What is the out-of-sample predictive performance of the BII total species abundance model? What are the key limitations of the model? I find that the model does not generalize well to unseen data, caused by the random effects structure of the linear mixed model. This structure reflects the study-block structure of the PREDICTS data used for training, which means that a lot of information is captured in parameters that cannot be used when predicting for new sites. Predictions made using the fixed effects are essentially flat, with very little linear relationship between observed and predicted values. Furthermore, the scaling method used to derive final index outputs from the model leads to a problematic distribution of output values. Going beyond the abundance model, the approach of combining these outputs with the compositional similarity model also seems questionable. The total species abundance metric is sensitive to abundant outlier species and does not reflect distribution evenness, which is typically a desired property of biodiversity metrics. These findings suggest caution in the use of the current BII implementation for policy and decision-making.

How does the predictive performance of the Bayesian hierarchical model compare to the BII? What proportion of improvements is driven by differences in model structure and differences in model covariates, respectively? The Bayesian hierarchical model achieves a 33% lower mean absolute error and a 52% lower median absolute error compared to the original BII model. The Bayesian model uses a hierarchical structure based on fundamental biogeographic factors, with additional bioclimatic and topographic covariates. Although this model has a lower R^2 of 0.485, compared to 0.620 of the BII baseline model, it captures a substantial portion of the explained variance previously attributed to the study-block random effects. The influence of the new hierarchical structure seems larger than the influence of additional covariates on the improved results. One problem with the hierarchical base model is that predictions are underbiased for large observed values and vice versa. Calibration of this model, using a simple polynomial regression model, leads to improvements of 58% and 68% over the BII baseline, respectively. Although the calibration step is somewhat experimental, it demonstrates the importance of

calibrating regression models.

What are the differences in final index outputs between the models? On average, the predicted abundance values of the hierarchical base model (before calibration) are 26% lower, and the final index outputs are 21% lower, than the BII baseline model. However, given the issues with the scaling method highlighted above, it is hard to say anything about the meaning of these differences. The main takeaway is that more work is required on model structure to obtain results that are consistent and sensible.

In summary, there are three important contributions of this study. First, it demonstrates the effectiveness and potential of using Bayesian hierarchical models for complex biodiversity modeling at scale. Second, it is possibly the first real attempt to systematically evaluate a global biodiversity model in terms of predictive performance, thereby starting to fill a key gap in the current literature. Third, it highlights some issues with the current BII implementation, which is one of the most well-known models in this domain, used by IPBES and GEO BON as a core indicator. Combining the best elements of existing models, such as BII and GLOBIO, with the strengths of Bayesian hierarchical models is a promising avenue for future research in this area. Together with advancements in satellite data and other technologies, accurate and robust biodiversity models can provide the foundation for near-real time monitoring of the health of ecosystems. This is important for informing policy and decisions by actors in the public and private sectors, to promote nature-positive outcomes.

References

[1] S. Naeem, J. E. Duffy, and E. Zavaleta, "The Functions of Biological Diversity in an Age of Extinction," *Science*, vol. 336, pp. 1401–1406, Jun. 2012.

[2] B. J. Cardinale *et al.*, "Biodiversity loss and its impact on humanity," *Nature*, vol. 486, pp. 59–67, Jun. 2012.

[3] D. Leclère *et al.*, "Bending the curve of terrestrial biodiversity needs an integrated strategy," *Nature*, vol. 585, pp. 551–556, Sep. 2020.

[4] IPBES, "Global assessment report on biodiversity and ecosystem services," Bonn, Germany, 2019.

[5] P. Jaureguiberry *et al.*, "The direct drivers of recent global anthropogenic biodiversity loss," *Science Advances*, vol. 8, Nov. 2022.

[6] F. E. B. Spooner, R. G. Pearson, and R. Freeman, "Rapid warming is associated with population decline among terrestrial birds and mammals globally," *Global Change Biology*, vol. 24, pp. 4521–4531, Oct. 2018.

[7] International Union for Conservation of Nature, "The IUCN Red List of Threatened Species. Version 2023-1," 2023, accessed on: 2024-02-08.

[8] S. E. H. Ledger *et al.*, "Past, present, and future of the Living Planet Index," *npj Biodiversity*, vol. 2, Jun. 2023.

[9] A. M. Schipper *et al.*, "Projecting terrestrial biodiversity intactness with GLOBIO 4," *Global Change Biology*, vol. 26, pp. 760–771, Feb. 2020.

[10] R. J. Scholes and R. Biggs, "A biodiversity intactness index," *Nature*, vol. 434, pp. 45–49, Mar. 2005.

[11] T. Newbold *et al.*, "Global effects of land use on local terrestrial biodiversity," *Nature*, vol. 520, pp. 45–50, Apr. 2015.

[12] J. P. G. Jones *et al.*, "The Why, What, and How of Global Biodiversity Indicators Beyond the 2010 Target," *Conservation Biology*, vol. 25, no. 3, pp. 450–457, Jun. 2010.

[13] L. N. Hudson *et al.*, "The PREDICTS database: a global database of how local terrestrial biodiversity responds to human impacts," *Ecology and Evolution*, vol. 4, pp. 4701–4735, Dec. 2014.

[14] E. Olsen, G. Fay, S. Gaichas, R. Gamble, S. Lucey, and J. S. Link, "Ecosystem Model Skill Assessment. Yes We Can!" *PLoS ONE*, vol. 11, Jan. 2016.

[15] K. E. Watermeyer *et al.*, "Using Decision Science to Evaluate Global Biodiversity Indices," *Conservation Biology*, vol. 35, pp. 492–501, Apr. 2020.

[16] R. M. Dorazio, "Bayesian data analysis in population ecology: motivations, methods, and benefits," *Population Ecology*, vol. 58, pp. 31–44, Jan. 2016.

[17] I. Secretariat, "Biodiversity Intactness Index: Fact Sheet for IPBES Assessments," 2017, accessed 2024-04-26.

[18] A. De Palma *et al.*, "Annual changes in the Biodiversity Intactness Index in tropical and subtropical forest biomes, 2001–2012," *Scientific Reports*, vol. 11, no. 20249, Oct. 2021.

[19] Convention on Biological Diversity, "Article 2. use of terms," 1993, accessed on: 2024-01-19.

[20] C. Duncan, J. Thompson, and N. Pettorelli, "The quest for a mechanistic understanding of biodiversity – ecosystem services relationships," *Proceedings of the Royal Society B*, vol. 282, no. 20151348, Oct. 2015.

[21] O. Bastian, "The role of biodiversity in supporting ecosystem services in Natura 2000 sites," *Ecological Indicators*, vol. 24, pp. 12–22, Jan. 2013.

[22] G. M. Mace, K. Norris, and A. H. Fitter, "Biodiversity and ecosystem services: a multilayered relationship," *Trends in Ecology & Evolution*, vol. 27, pp. 19–26, Jan. 2012.

[23] T. H. Oliver *et al.*, "Biodiversity and Resilience of Ecosystem Functions," *Trends in Ecology Evolution*, vol. 30, pp. 673–684, Oct. 2015.

[24] E. S. Zavaleta, J. R. Pasari, K. B. Hulvey, and G. D. Tilman, "Sustaining multiple ecosystem functions in grassland communities requires higher biodiversity," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 107, no. 4, pp. 1443–1446, Jan. 2010.

[25] G. Peterson, C. R. Allen, and C. S. Holling, "Ecological Resilience, Biodiversity, and Scale," *Ecosystems*, vol. 1, pp. 6–18, Jan. 1998.

[26] T. Elmqvist *et al.*, "Response diversity, ecosystem change, and resilience," *Frontiers in Ecology and the Environment*, vol. 1, pp. 488–494, Nov. 2003.

[27] S. Hoban *et al.*, "Global Commitments to Conserving and Monitoring Genetic Diversity Are Now Necessary and Feasible," *BioScience*, vol. 71, pp. 964–976, Sep. 2021.

[28] M. O. Hill, "Diversity and Evenness: A Unifying Notation and Its Consequences," *Ecology*, vol. 54, pp. 427–432, Mar. 1973.

[29] L. Jost, "Partitioning Diversity into Independent Alpha and Beta Components," *Ecology*, vol. 88, pp. 2427–2439, Oct. 2007.

[30] T. Newbold *et al.*, "Has land use pushed terrestrial biodiversity beyond the planetary boundary? A global assessment," *Science*, vol. 353, pp. 288–291, Jul. 2016.

[31] K. Richardson *et al.*, "Earth Beyond Six of Nine Planetary Boundaries," *Science Advances*, vol. 9, no. 37, Sep. 2023.

[32] P. A. Martin, R. E. Green, and A. Balmford, "The Biodiversity Intactness Index May Underestimate Losses," *Nature Ecology & Evolution*, vol. 3, pp. 862–863, May 2019.

[33] A. De Palma *et al.*, "Annual changes in the Biodiversity Intactness Index in tropical and subtropical forest biomes, 2001-2012: Supplementary Material 1," Oct. 2021.

[34] S. Nakagawa and H. Schielzeth, "A general and simple method for obtaining R2 from generalized linear mixed-effects models," *Methods in Ecology and Evolution*, vol. 4, pp. 133–142, Feb. 2013.

[35] M. Dornelas *et al.*, "BioTIME: A database of biodiversity time series for the Anthropocene," *Global Ecology and Biogeography*, vol. 27, no. 7, pp. 760–786, Jul. 2018.

[36] J. R. Sauer *et al.*, "The first 50 years of the North American Breeding Bird Survey," *The Condor*, vol. 119, pp. 576–593, Jul. 2017.

[37] T. F. Johnson *et al.*, "Revealing uncertainty in the status of biodiversity change," *Nature*, Mar. 2024.

[38] O. Venter *et al.*, "Sixteen years of change in the global terrestrial human footprint and implications for biodiversity conservation," *Nature Communications*, vol. 7, no. 12558, Aug. 2016.

[39] M. Di Marco, O. Venter, H. P. Possingham, and J. E. M. Watson, "Changes in human footprint drive changes in species extinction risk," *Nature Communications*, vol. 9, no. 4621, Nov. 2018.

[40] K.-H. Erb *et al.*, "Unexpectedly large impact of forest management and grazing on global vegetation biomass," *Nature*, vol. 553, pp. 73–76, Dec. 2018.

[41] H. S. Grantham *et al.*, "Anthropogenic modification of forests means only 40% of remaining forests have high ecosystem integrity," *Nature Communications*, vol. 11, no. 5978, Dec. 2020.

[42] X. A. Harrison *et al.*, "A brief introduction to mixed effects modelling and multi-model inference in ecology," *PeerJ*, vol. 6, p. e4794, Jan. 2018.

[43] J. C. Pinheiro and D. M. Bates, *Linear Mixed-Effects Models: Basic Concepts and Examples.* New York, NY: Springer, 2000.

[44] A. Gelman, "Multilevel (Hierarchical) Modeling: What It Can and Cannot Do," *Technometrics*, vol. 48, no. 3, Dec. 2006.

[45] D. J. Barr, R. Levy, C. Scheepers, and H. J. Tily, "Random Effects Structure for Confirmatory Hypothesis Testing: Keep It Maximal," *Journal of Memory and Language*, vol. 68, no. 3, pp. 255–278, Apr. 2013.

[46] J. Jiang and T. Nguyen, *Linear and Generalized Linear Mixed Models and Their Applications*, 2nd ed., ser. Springer Series in Statistics. Portland, OR, USA: Springer, 2021.

[47] H. Schielzeth and S. Nakagawa, "Nested by Design: Model Fitting and Interpretation in a Mixed Model Era," *Methods in Ecology & Evolution*, vol. 4, no. 1, pp. 14–24, Jan. 2013.

[48] X. A. Harrison, "A Comparison of Observation-Level Random Effect and Beta-Binomial Models for Modelling Overdispersion in Binomial Data in Ecology & Evolution," *PeerJ*, vol. 3, p. e1114, Jul. 2015.

[49] A. Gelman, J. Carlin, H. Stern, D. Dunson, A. Vehtari, and D. Rubin, *Bayesian Data Analysis*, 3rd ed. Columbia University, 2020.

[50] A. Gelman, J. Hill, and M. Yajima, "Why We (Usually) Don't Have to Worry About Multiple Comparisons," *Journal of Research on Educational Effectiveness*, vol. 5, no. 2, Apr. 2012.

[51] N. P. Lemoine, "Moving Beyond Noninformative Priors: Why and How to Choose Weakly Informative Priors in Bayesian Analyses," *OIKOS*, vol. 128, no. 7, pp. 912–928, Jul. 2019.

[52] A. Gelman, D. Simpson, and M. Betancourt, "The Prior Can Often Only Be Understood in the Context of the Likelihood," *Entropy*, vol. 19, p. 555, Oct. 2017.

[53] J. Gabry, D. Simpson, A. Vehtari, M. Betancourt, and A. Gelman, "Visualization in Bayesian Workflow," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 182, no. 2, pp. 389–402, Feb. 2019.

[54] A. Gelman, D. Lee, and J. Guo, "Stan: A Probabilistic Programming Language for Bayesian Inference and Optimization," *Journal of Educational and Behavioral Statistics*, vol. 40, no. 5, Oct. 2015.

[55] A. Gelman, F. Bois, and J. Jiang, "Physiological Pharmacokinetic Analysis Using Population Modeling and Informative Prior Distributions," *Journal of the American Statistical Association*, vol. 91, no. 436, Feb. 1996.

[56] Stan Development Team, *Stan User's Guide*, 2023, accessed on: 2024-05-22.

[57] S. W. Nydick, "The Wishart and Inverse Wishart Distributions," *Electronic Journal of Statistics*, vol. 6, pp. 1–19, May 2012.

[58] B. M. Bolker *et al.*, "Generalized linear mixed models: a practical guide for ecology and evolution," *Trends in Ecology & Evolution*, vol. 24, no. 3, Mar. 2009.

[59] J. A. Nelder and R. W. M. Wedderburn, "Generalized Linear Models," *Royal Statistical Society Journal Series A: General*, vol. 135, pp. 370–384, 1972.

[60] S. Wright, "Correlation and causation," *Journal of Agricultural Research*, vol. 20, no. 7, pp. 557–585, 1921.

[61] N. Draper and H. Smith, *Applied Regression Analysis*, 3rd ed. New York: Wiley, 1998.

[62] P. C. D. Johnson, "Extension of Nakagawa & Schielzeth's R2GLMM to random slopes models," *Methods in Ecology and Evolution*, vol. 5, pp. 944–946, Sep. 2014.

[63] S. Nakagawa, P. C. D. Johnson, and H. Schielzeth, "The Coefficient of Determination R^2 and Intra-class Correlation Coefficient from Generalized Linear Mixed-Effects Models Revisited and Expanded," *Journal of the Royal Society Interface*, vol. 14, no. 20170213, Sep. 2017.

[64] B. C. Jaeger, L. J. Edwards, K. Das, and P. K. Sen, "An R^2 statistic for fixed effects in the generalized linear mixed model," *Journal of Applied Statistics*, vol. 44, no. 6, Dec. 2017.

[65] A. Gelman, B. Goodrich, J. Gabry, and A. Vehtari, "R-squared for Bayesian regression models," *The American Statistician*, vol. 73, no. 3, May 2019.

[66] H. Akaike, "Information theory as an extension of the maximum likelihood principle," in *2nd International Symposium on Information Theory*, B. Petrov and F. Csaki, Eds. Budapest: Akademiai Kiado, 1973, pp. 267–281.

[67] G. Schwarz, "Estimating the dimension of a model," *Annals of Statistics*, vol. 6, pp. 461–464, Mar. 1978.

[68] D. Spiegelhalter, N. Best, B. Carlin, and A. van der Linde, "Bayesian measures of model complexity and fit," *Journal of the Royal Statistical Society Series B-Statistical Methodology*, vol. 64, pp. 583–616, Oct. 2002.

[69] A. Vehtari, A. Gelman, and J. Gabry, "Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC," *Statistics and Computing*, vol. 27, pp. 1413–1432, Aug. 2017.

[70] C. Sammut and G. I. Webb, "Mean Squared Error," in *Encyclopedia of Machine Learning*. Springer, 2010, p. 653.

[71] T. Chai and R. R. Draxler, "Root Mean Square Error (RMSE) or Mean Absolute Error (MAE)?" *Geoscientific Model Development Discussions*, vol. 7, pp. 1525–1534, 2014.

[72] C. Sammut and G. I. Webb, "Mean Absolute Error," in *Encyclopedia of Machine Learning*, C. Sammut and G. I. Webb, Eds. Springer, 2010, p. 652.

[73] A. de Myttenaere, B. Golden, B. L. Grand, and F. Rossi, "Mean Absolute Percentage Error for Regression Models," *Neurocomputing*, vol. 192, pp. 38–48, Jun. 2016.

[74] D. Chicco, M. J. Warrens, and G. Jurman, "The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation," *PeerJ Computer Science*, Jul. 2021.

[75] L. N. Hudson *et al.*, "The database of the PREDICTS (Projecting Responses of Ecological Diversity In Changing Terrestrial Systems) project," *Ecology and Evolution*, vol. 7, pp. 145–188, Jan. 2017.

[76] L. Hudson *et al.*, "The 2016 Release of the PREDICTS Database V1.1," 2023, accessed on: 2024-02-13.

[77] S. Contu *et al.*, "Release of Data Added to the PREDICTS Database (November 2022)," 2022, accessed on: 2024-02-13.

[78] A. De Palma *et al.*, "Calculating the Biodiversity Intactness Index: the PREDICTS implementation," Nov. 2021, accessed: 2023-11-19.

[79] E. Doxsey-Whitfield *et al.*, "Taking Advantage of the Improved Availability of Census Data: A First Look at the Gridded Population of the World, Version 4," *Papers in Applied Geography*, vol. 1, pp. 226–234, Jul. 2015.

[80] Center for International Earth Science Information Network (CIESIN), Columbia University, "Gridded Population of the World, Version 4 (GPWv4): Population Density Adjusted to Match 2015 Revision of UN WPP Country Totals, Revision 11," NASA Socioeconomic Data and Applications Center (SEDAC), 2018, accessed on: 2024-01-30.

[81] Center for International Earth Science Information Network (CIESIN), Columbia University, and Information Technology Outreach Services (ITOS), University of Georgia, "Global Roads Open Access Data Set, Version 1 (gROADSv1)," Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC), 2013, accessed on: 2024-01-30.

[82] S. E. Fick and R. J. Hijmans, "WorldClim 2: new 1km spatial resolution climate surfaces for global land areas," *International Journal of Climatology*, vol. 37, pp. 4302–4315, May 2017.

[83] G. Amatulli, S. Domisch, M.-N. Tuanmu, B. Parmentier, A. Ranipeta, J. Malczyk, and W. Jetz, "A Suite of Global, Cross-Scale Topographic Variables for Environmental and Biodiversity Modeling," *Scientific Data*, vol. 5, 2018.

[84] E. O. Box and K. Fujiwara, *Vegetation Types and Their Broad-Scale Distribution.* Oxford: Blackwell Scientific, 2005, pp. 106–128.

[85] W. D. Bowman and S. D. Hacker, *Ecology*, 5th ed. Oxford University Press, 2021.

[86] D. M. Olson and E. Dinerstein, "The Global 200: A Representation Approach to Conserving the Earth's Most Biologically Valuable Ecoregions," *Conservation Biology*, vol. 12, pp. 502–515, Jun. 1998.

[87] R. G. Bailey, *Ecoregions: The Ecosystem Geography of the Oceans and Continents*, 2nd ed. Springer, 2014.

[88] P. Li, "Efficient MCMC Estimation of Inflated Beta Regression Models," *Computational Statistics*, vol. 33, no. 1, pp. 127–158, Mar. 2018.

[89] R. A. Stine, "Statistics 622: Module 4 Calibration," Nov. 2011.

[90] E. Jolly, "Pymer4: Connecting R and Python for Linear Mixed Modeling," *Journal of Open Source Software*, vol. 3, no. 31, p. 862, Nov. 2018.

[91] O. Abril-Pla *et al.*, "PyMC: A Modern, and Comprehensive Probabilistic Programming Framework in Python," *PeerJ Computer Science*, vol. 9, p. e1516, Sep. 2023.

[92] M. D. Hoffman and A. Gelman, "The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo," *Journal of Machine Learning Research*, vol. 15, pp. 1351–1381, Apr. 2014.

[93] D. Phan, N. Pradhan, and M. Jankowiak, "Composable Effects for Flexible and Accelerated Probabilistic Programming in NumPyro," *arXiv preprint*, Dec. 2019.

[94] E. Bingham *et al.*, "Pyro: Deep Universal Probabilistic Programming," *Journal of Machine Learning Research*, vol. 20, pp. 28:1–28:6, Feb. 2019.

[95] J. Bradbury *et al.*, "JAX: Composable Transformations of Python+NumPy Programs," 2018.

[96] Planet Labs PBC, "Accelerating Biodiversity and Ecosystem Reporting: Feasible and Cost-effective Reporting with AI, Earth Observation, & Ecosystem Science," White Paper, 2024.

[97] The World Bank DataBank, "Institutional Quality, World Governance Indicators," 2024, accessed: 2024-05-04.

[98] Organisation for Economic Co-operation and Development (OECD), "Environmental Policy Stringency Index," 2024, accessed: 2024-05-04.

A Supplementary materials

This appendix contains supporting materials referenced in the main text, and some additional tables with parameter estimates.

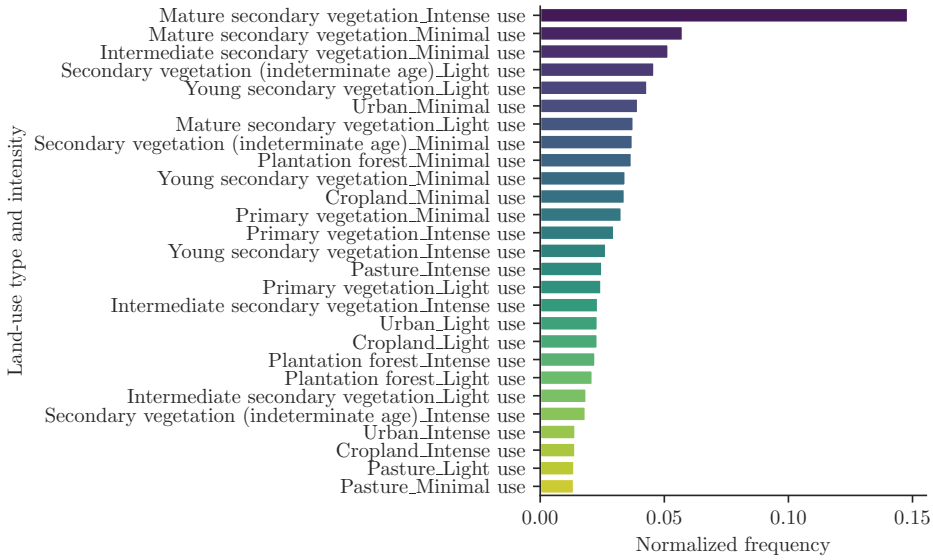


FIGURE 14: The number of times each land-use type and intensity combination is the highest abundance site within a study, adjusted for the number of such sites in total in the data.

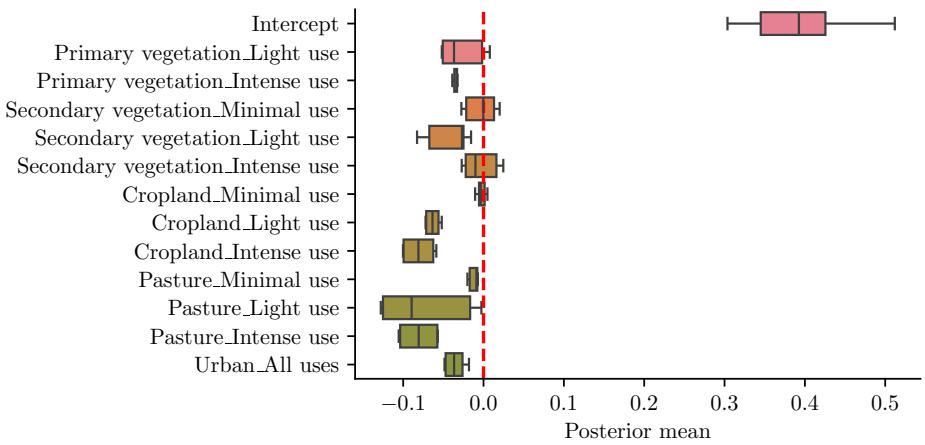


FIGURE 15: Land-use parameters at realm level from Bayesian hierarchical model. Based on model without square-root transformation, to make parameters more interpretable.

Table 5: Parameter estimates of BII baseline model (study slope model), before back-transformation

Variable	Est	2.5% CI	97.5% CI	P-val	Sig
Intercept	0.669	0.637	0.701	0.000	***
Primary vegetation Light use	-0.045	-0.085	-0.006	0.026	*
Primary vegetation Intense use	-0.037	-0.098	0.024	0.255	
Secondary vegetation Minimal use	-0.018	-0.052	0.016	0.302	
Secondary vegetation Light use	-0.038	-0.074	-0.003	0.039	*
Secondary vegetation Intense use	-0.071	-0.115	-0.027	0.002	**
Cropland Minimal use	-0.010	-0.070	0.049	0.736	
Cropland Light Intense	-0.161	-0.251	-0.072	0.002	**
Pasture Minimal use	-0.041	-0.112	0.029	0.253	
Pasture Light Intense	-0.090	-0.182	0.001	0.060	.
Urban All uses	-0.027	-0.140	0.086	0.645	
Pop density 1km log	0.001	-0.023	0.025	0.928	
Road density 50km cbrt	0.007	-0.022	0.036	0.628	
Mean pop density 1km log	-0.020	-0.048	0.008	0.169	
Primary vegetation Light use x Pop density 1km log	-0.012	-0.043	0.018	0.423	
Primary vegetation Intense use x Pop density 1km log	-0.007	-0.061	0.046	0.790	
Secondary vegetation Minimal use x Pop density 1km log	-0.018	-0.044	0.008	0.183	
Secondary vegetation Light use x Pop density 1km log	-0.004	-0.034	0.025	0.765	
Secondary vegetation Intense use x Pop density 1km log	-0.008	-0.038	0.021	0.580	
Cropland Minimal use x Pop density 1km log	0.034	-0.020	0.089	0.225	
Cropland Light Intense x Pop density 1km log	-0.006	-0.078	0.065	0.860	
Pasture Minimal use x Pop density 1km log	0.041	-0.020	0.101	0.196	
Pasture Light Intense x Pop density 1km log	0.019	-0.044	0.081	0.560	
Urban All uses x Pop density 1km log	-0.013	-0.067	0.041	0.637	
Primary vegetation Light use x Road density 50km cbrt	-0.013	-0.053	0.026	0.512	
Primary vegetation Intense use x Road density 50km cbrt	0.038	-0.018	0.095	0.189	
Secondary vegetation Minimal use x Road density 50km cbrt	0.009	-0.021	0.038	0.573	
Secondary vegetation Light use x Road density 50km cbrt	-0.019	-0.052	0.015	0.278	
Secondary vegetation Intense use x Road density 50km cbrt	-0.033	-0.074	0.007	0.108	
Cropland Minimal use x Road density 50km cbrt	-0.009	-0.071	0.053	0.768	
Cropland Light Intense x Road density 50km cbrt	-0.073	-0.151	0.005	0.076	.
Pasture Minimal use x Road density 50km cbrt	0.079	-0.016	0.174	0.108	
Pasture Light Intense x Road density 50km cbrt	0.029	-0.053	0.110	0.494	
Urban All uses x Road density 50km cbrt	0.036	-0.066	0.137	0.496	

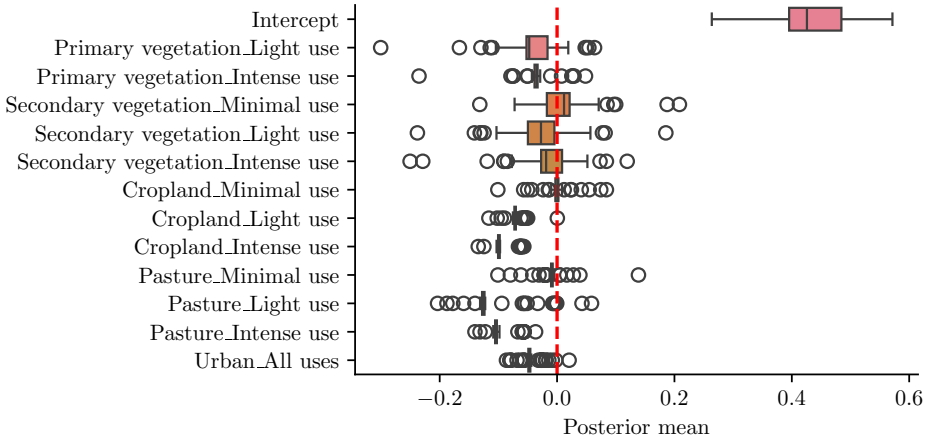


FIGURE 16: Land-use parameters at ecoregion level from Bayesian hierarchical model. Based on model without square-root transformation, to make parameters more interpretable.

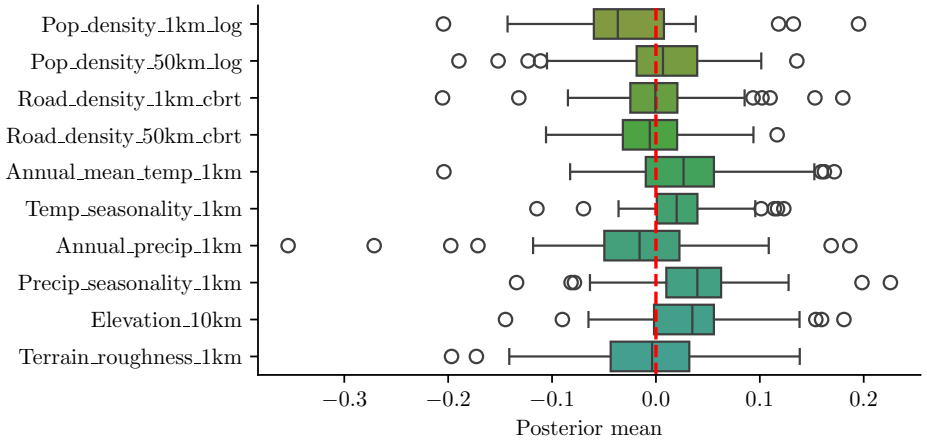


FIGURE 17: Population density, road density, bioclimatic and topographic parameters at ecoregion level from Bayesian hierarchical model. Based on model without square-root transformation, to make parameters more interpretable.

Table 6: Biome-level parameters of Bayesian hierarchical model (granular LUI), after back-transformation

Variable	Dry broadl.	Moist broadl.	Coniferous
Intercept	0.446	0.356	0.374
Primary vegetation Light use	-0.050	0.006	-0.025
Primary vegetation Intense use	-0.036	-0.033	-0.034
Secondary vegetation Minimal use	0.014	-0.025	-0.013
Secondary vegetation Light use	-0.025	-0.080	-0.025
Secondary vegetation Intense use	-0.022	0.023	-0.002
Cropland Minimal use	0.002	-0.005	-0.010
Cropland Light use	-0.071	-0.056	-0.052
Cropland Intense use	-0.099	-0.063	-0.059
Pasture Minimal use	-0.009	-0.020	-0.007
Pasture Light use	-0.125	-0.004	-0.054
Pasture Intense use	-0.104	-0.057	-0.057
Urban All uses	-0.047	-0.026	-0.019
Pop density 1km log	-0.029	-0.008	-0.016
Pop density 50km log	0.002	0.006	-0.004
Road density 1km cbrt	-0.001	-0.006	0.000
Road density 50km cbrt	-0.008	0.016	0.007
Annual mean temp 1km	0.025	0.031	0.070
Temp seasonality 1km	0.019	0.049	0.015
Annual precip 1km	-0.015	-0.044	0.010
Precip seasonality 1km	0.038	0.010	0.013
Elevation 10km	0.032	0.023	0.036
Terrain roughness 1km	-0.010	0.009	0.000
Primary vegetation Light use x Pop density 50km log	-0.010	-0.051	-0.024
Primary vegetation Intense use x Pop density 50km log	0.038	0.022	0.022
Secondary vegetation Minimal use x Pop density 50km log	-0.017	-0.032	-0.013
Secondary vegetation Light use x Pop density 50km log	0.015	0.038	0.021
Secondary vegetation Intense use x Pop density 50km log	-0.055	-0.074	-0.049
Cropland Minimal use x Pop density 50km log	0.053	0.035	0.029
Cropland Light use x Pop density 50km log	0.108	0.079	0.070
Cropland Intense use x Pop density 50km log	0.022	0.033	0.020
Pasture Minimal use x Pop density 50km log	-0.040	-0.037	-0.029
Pasture Light use x Pop density 50km log	-0.104	-0.039	-0.061
Pasture Intense use x Pop density 50km log	-0.040	-0.028	-0.026
Urban All uses x Pop density 50km log	-0.049	-0.032	-0.028
Primary vegetation Light use x Road density 50km cbrt	-0.002	-0.039	-0.021
Primary vegetation Intense use x Road density 50km cbrt	-0.022	-0.028	-0.017
Secondary vegetation Minimal use x Road density 50km cbrt	0.022	-0.033	0.007
Secondary vegetation Light use x Road density 50km cbrt	-0.028	-0.011	-0.047
Secondary vegetation Intense use x Road density 50km cbrt	-0.016	-0.022	-0.018
Cropland Minimal use x Road density 50km cbrt	-0.024	-0.013	-0.008
Cropland Light use x Road density 50km cbrt	-0.016	-0.036	-0.017
Cropland Intense use x Road density 50km cbrt	0.024	0.010	0.012
Pasture Minimal use x Road density 50km cbrt	0.071	0.032	0.041
Pasture Light use x Road density 50km cbrt	0.014	-0.056	-0.026
Pasture Intense use x Road density 50km cbrt	0.054	0.026	0.033
Urban All uses x Road density 50km cbrt	0.041	0.039	0.056

Table 7: Population-level parameters of Bayesian hierarchical model (granular LUI), after back-transformation

Variable	Mean	SD	HDI 3%	HDI 97%
Intercept	0.419	0.063	0.301	0.537
Primary vegetation Light use	-0.022	0.048	-0.107	0.074
Primary vegetation Intense use	-0.030	0.055	-0.137	0.070
Secondary vegetation Minimal use	-0.004	0.048	-0.098	0.085
Secondary vegetation Light use	-0.036	0.049	-0.134	0.054
Secondary vegetation Intense use	-0.004	0.052	-0.102	0.092
Cropland Minimal use	-0.004	0.056	-0.111	0.099
Cropland Light use	-0.052	0.068	-0.177	0.080
Cropland Intense use	-0.060	0.088	-0.220	0.107
Pasture Minimal use	-0.011	0.057	-0.116	0.095
Pasture Light use	-0.051	0.068	-0.184	0.072
Pasture Intense use	-0.059	0.082	-0.214	0.092
Urban All uses	-0.028	0.070	-0.160	0.106
Pop density 1km log	-0.018	0.037	-0.082	0.064
Pop density 50km log	0.002	0.046	-0.091	0.085
Road density 1km cbrt	-0.003	0.031	-0.061	0.055
Road density 50km cbrt	0.003	0.045	-0.076	0.100
Annual mean temp 1km	0.036	0.043	-0.046	0.117
Temp seasonality 1km	0.025	0.045	-0.058	0.115
Annual precip 1km	-0.015	0.050	-0.114	0.078
Precip seasonality 1km	0.019	0.049	-0.074	0.113
Elevation 10km	0.028	0.045	-0.059	0.114
Terrain roughness 1km	-0.000	0.034	-0.065	0.067
Primary vegetation Light use x Pop density 50km log	-0.024	0.051	-0.125	0.069
Primary vegetation Intense use x Pop density 50km log	0.025	0.057	-0.079	0.129
Secondary vegetation Minimal use x Pop density 50km log	-0.017	0.051	-0.113	0.085
Secondary vegetation Light use x Pop density 50km log	0.021	0.055	-0.086	0.131
Secondary vegetation Intense use x Pop density 50km log	-0.050	0.059	-0.161	0.065
Cropland Minimal use x Pop density 50km log	0.034	0.064	-0.089	0.151
Cropland Light use x Pop density 50km log	0.070	0.084	-0.091	0.219
Cropland Intense use x Pop density 50km log	0.019	0.091	-0.151	0.185
Pasture Minimal use x Pop density 50km log	-0.031	0.062	-0.143	0.090
Pasture Light use x Pop density 50km log	-0.057	0.079	-0.209	0.090
Pasture Intense use x Pop density 50km log	-0.026	0.079	-0.169	0.126
Urban All uses x Pop density 50km log	-0.031	0.069	-0.156	0.096
Primary vegetation Light use x Road density 50km cbrt	-0.018	0.046	-0.114	0.063
Primary vegetation Intense use x Road density 50km cbrt	-0.019	0.058	-0.123	0.092
Secondary vegetation Minimal use x Road density 50km cbrt	-0.001	0.052	-0.102	0.094
Secondary vegetation Light use x Road density 50km cbrt	-0.025	0.051	-0.127	0.069
Secondary vegetation Intense use x Road density 50km cbrt	-0.016	0.050	-0.113	0.079
Cropland Minimal use x Road density 50km cbrt	-0.013	0.062	-0.126	0.102
Cropland Light use x Road density 50km cbrt	-0.020	0.066	-0.138	0.113
Cropland Intense use x Road density 50km cbrt	0.013	0.081	-0.141	0.155
Pasture Minimal use x Road density 50km cbrt	0.042	0.071	-0.091	0.174
Pasture Light use x Road density 50km cbrt	-0.016	0.070	-0.146	0.116
Pasture Intense use x Road density 50km cbrt	0.032	0.074	-0.114	0.165
Urban All uses x Road density 50km cbrt	0.039	0.070	-0.086	0.179